

Lecture1

Optimization

Introduction, Basic concepts

Vector Spaces and Matrices

We define a *column* n -vector to be an array of n numbers, denoted

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}.$$

The number a_i is called the i th component of the vector $\mathbf{a}$. Denote by $\mathbb{R}$ the set of real numbers, and by $\mathbb{R}^n$ the set of column n -vectors with real components. We call $\mathbb{R}^n$ an n -dimensional *real vector space*. We commonly denote elements of $\mathbb{R}^n$ by lower-case bold letters, e.g., $\mathbf{x}$. The components of $\mathbf{x} \in \mathbb{R}^n$ are denoted $x_1, \dots, x_n$.

We define a *row* n -vector as

$$[a_1, a_2, \dots, a_n].$$

The *transpose* of a given column vector $\mathbf{a}$ is a row vector with corresponding elements, denoted $\mathbf{a}^T$. Equivalently, we may write $\mathbf{a} = [a_1, a_2, \dots, a_n]^T$.

In this course, the term vector (without the qualifier row or column) will be refer to “column vector”.

A set of vectors $\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$ is said to be *linearly independent* if the equality

$$\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k = \mathbf{0}$$

implies that all coefficients $\alpha_i, i = 1, \dots, k$, are equal to zero. A set of the vectors $\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$ is *linearly dependent* if it is not linearly independent.

❖ What about the set composed of the single vector $\mathbf{0}$?

Note that the set composed of the single vector $\mathbf{0}$ is linearly dependent, for if $\alpha \neq 0$ then $\alpha \mathbf{0} = \mathbf{0}$. In fact, any set of vectors containing the vector $\mathbf{0}$ is linearly dependent.

❖ What about a set composed of the single nonzero vector?

A set composed of a single nonzero vector $\mathbf{a} \neq \mathbf{0}$ is linearly independent since $\alpha \mathbf{a} = \mathbf{0}$ implies $\alpha = 0$.

A vector $\mathbf{a}$ is said to be a *linear combination* of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ if there are scalars $\alpha_1, \dots, \alpha_k$ such that

$$\mathbf{a} = \alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k.$$

Proposition 2.1 *A set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ is linearly dependent if and only if one of the vectors from the set is a linear combination of the remaining vectors.*

Proof. $\Rightarrow$: If $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ is linearly dependent then

$$\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k = \mathbf{0},$$

where at least one of the scalars $\alpha_i \neq 0$, whence

$$\mathbf{a}_i = -\frac{\alpha_1}{\alpha_i} \mathbf{a}_1 - \frac{\alpha_2}{\alpha_i} \mathbf{a}_2 - \dots - \frac{\alpha_k}{\alpha_i} \mathbf{a}_k.$$

$\Leftarrow$: Suppose

$$\mathbf{a}_1 = \alpha_2 \mathbf{a}_2 + \alpha_3 \mathbf{a}_3 + \dots + \alpha_k \mathbf{a}_k,$$

then

$$(-1)\mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k = \mathbf{0}.$$

Because the first scalar is nonzero, the set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ is linearly dependent. The same argument holds if $\mathbf{a}_i, i = 2, \dots, k$, is a linear combination of the remaining vectors. ■

A subset $\mathcal{V}$ of $\mathbb{R}^n$ is called a *subspace* of $\mathbb{R}^n$ if $\mathcal{V}$ is closed under the operations of vector addition and scalar multiplication. That is, if $\mathbf{a}$ and $\mathbf{b}$ are vectors in $\mathcal{V}$, then the vectors $\mathbf{a} + \mathbf{b}$ and $\alpha\mathbf{a}$ are also in $\mathcal{V}$ for every scalar α .

Every subspace contains the zero vector $\mathbf{0}$, for if $\mathbf{a}$ is an element of the subspace, so is $(-1)\mathbf{a} = -\mathbf{a}$. Hence, $\mathbf{a} - \mathbf{a} = \mathbf{0}$ also belongs to the subspace.

Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ be arbitrary vectors in $\mathbb{R}^n$. The set of all their linear combinations is called the *span* of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ and is denoted

$$\text{span}[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k] = \left\{ \sum_{i=1}^k \alpha_i \mathbf{a}_i : \alpha_1, \dots, \alpha_k \in \mathbb{R} \right\}.$$

Given a vector $\mathbf{a}$, the subspace $\text{span}[\mathbf{a}]$ is composed of the vectors $\alpha\mathbf{a}$, where α is an arbitrary real number ($\alpha \in \mathbb{R}$). Also observe that if $\mathbf{a}$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ then

$$\text{span}[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k, \mathbf{a}] = \text{span}[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k].$$

The span of any set of vectors is a subspace.

Given a subspace $\mathcal{V}$, any set of linearly independent vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\} \subset \mathcal{V}$ such that $\mathcal{V} = \text{span}[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k]$ is referred to as a *basis* of the subspace $\mathcal{V}$. All bases of a subspace $\mathcal{V}$ contain the same number of vectors. This number is called the *dimension* of $\mathcal{V}$, denoted $\dim \mathcal{V}$.

Proposition 2.2 *If $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ is a basis of $\mathcal{V}$, then any vector $\mathbf{a}$ of $\mathcal{V}$ can be represented uniquely as*

$$\mathbf{a} = \alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k,$$

where $\alpha_i \in \mathbb{R}$, $i = 1, 2, \dots, k$.

Proof. To prove the uniqueness of the representation of $\mathbf{a}$ in terms of the basis vectors, assume that

$$\mathbf{a} = \alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \cdots + \alpha_k \mathbf{a}_k$$

and

$$\mathbf{a} = \beta_1 \mathbf{a}_1 + \beta_2 \mathbf{a}_2 + \cdots + \beta_k \mathbf{a}_k.$$

We now show that $\alpha_i = \beta_i, i = 1, \dots, k$. We have

$$\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \cdots + \alpha_k \mathbf{a}_k = \beta_1 \mathbf{a}_1 + \beta_2 \mathbf{a}_2 + \cdots + \beta_k \mathbf{a}_k,$$

or

$$(\alpha_1 - \beta_1) \mathbf{a}_1 + (\alpha_2 - \beta_2) \mathbf{a}_2 + \cdots + (\alpha_k - \beta_k) \mathbf{a}_k = \mathbf{0}.$$

Because the set $\{\mathbf{a}_i : i = 1, 2, \dots, k\}$ is linearly independent, $\alpha_1 - \beta_1 = \alpha_2 - \beta_2 = \cdots = \alpha_k - \beta_k = 0$, that is, $\alpha_i = \beta_i, i = 1, \dots, k$. ■

Suppose we are given a basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ of $\mathcal{V}$ and a vector $\mathbf{a} \in \mathcal{V}$ such that

$$\mathbf{a} = \alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k.$$

The coefficients $\alpha_i, i = 1, \dots, k$, are called the *coordinates* of $\mathbf{a}$ with respect to the basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$.

The *natural basis* for $\mathbb{R}^n$ is the set of vectors

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

The reason for calling these vectors the natural basis is that

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n.$$

Rank of a matrix

A *matrix* is a rectangular array of numbers, commonly denoted by upper-case bold letters, e.g., $\mathbf{A}$. A matrix with m rows and n columns is called an $m \times n$ matrix, and we write

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \vdots & a_{m2} \\ \vdots & \cdots & \ddots & \cdots \\ a_{m1} & a_{2n} & \vdots & a_{mn} \end{bmatrix}.$$

Let us denote the k th column of $\mathbf{A}$ by $\mathbf{a}_k$, that is,

$$\mathbf{a}_k = \begin{bmatrix} a_{1k} \\ a_{2k} \\ \vdots \\ a_{mk} \end{bmatrix}.$$

The maximal number of linearly independent columns of $\mathbf{A}$ is called the *rank* of the matrix $\mathbf{A}$, denoted $\text{rank } \mathbf{A}$. Note that $\text{rank } \mathbf{A}$ is the dimension of $\text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n]$.

A p th-order *minor* of an $m \times n$ matrix $\mathbf{A}$, with $p \leq \min(m, n)$, is the determinant of a $p \times p$ matrix obtained from $\mathbf{A}$ by deleting $m - p$ rows and $n - p$ columns.

If a matrix $\mathbf{A}$ has an r th-order minor $|\mathbf{M}|$ with the properties (i) $|\mathbf{M}| \neq 0$ and (ii) any minor of $\mathbf{A}$ that is formed by adding a row and a column of $\mathbf{A}$ to $\mathbf{M}$ is zero, then

$$\text{rank } \mathbf{A} = r.$$

Thus, the rank of a matrix is equal to the highest order of its nonzero minor(s).

A *nonsingular* (or *invertible*) matrix is a square matrix whose determinant is nonzero.

Suppose that $\mathbf{A}$ is an $n \times n$ square matrix. Then, $\mathbf{A}$ is nonsingular if and only if there is another $n \times n$ matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n,$$

where $\mathbf{I}_n$ denotes the $n \times n$ *identity matrix*:

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

We call the above matrix $\mathbf{B}$ the *inverse matrix* of $\mathbf{A}$, and write $\mathbf{B} = \mathbf{A}^{-1}$.

Consider the $m \times n$ matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The *transpose* of $\mathbf{A}$, denoted $\mathbf{A}^T$, is the $n \times m$ matrix

$$\mathbf{A}^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix},$$

that is, the columns of $\mathbf{A}$ are the rows of $\mathbf{A}^T$, and vice versa. A matrix $\mathbf{A}$ is *symmetric* if $\mathbf{A} = \mathbf{A}^T$.

Inner Products and Norms

The absolute value of a real number a , denoted $|a|$, is defined as

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}.$$

The following formulas hold:

1. $|a| = |-a|$;
2. $-|a| \leq a \leq |a|$;
3. $|a + b| \leq |a| + |b|$;
4. $||a| - |b|| \leq |a - b| \leq |a| + |b|$;
5. $|ab| = |a||b|$;
6. $|a| \leq c$ and $|b| \leq d$ imply $|a + b| \leq c + d$;
7. The inequality $|a| < b$ is equivalent to $-b < a < b$ (i.e., $a < b$ and $-a < b$).
The same holds if we replace every occurrence of “<” by “ $\leq$.”
8. The inequality $|a| > b$ is equivalent to $a > b$ or $-a > b$. The same holds if we replace every occurrence of “>” by “ $\geq$.”

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we define the *Euclidean inner product* by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i = \mathbf{x}^T \mathbf{y}.$$

The inner product is a real-valued function $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ having the following properties:

1. Positivity: $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$;
2. Symmetry: $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
3. Additivity: $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$;
4. Homogeneity: $\langle r\mathbf{x}, \mathbf{y} \rangle = r\langle \mathbf{x}, \mathbf{y} \rangle$ for every $r \in \mathbb{R}$.

The properties of additivity and homogeneity in the second vector also hold, that is,

$$\begin{aligned}\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle &= \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle, \\ \langle \mathbf{x}, r\mathbf{y} \rangle &= r\langle \mathbf{x}, \mathbf{y} \rangle \quad \text{for every } r \in \mathbb{R}.\end{aligned}$$

The vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be *orthogonal* if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

The *Euclidean norm* of a vector $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{\mathbf{x}^T \mathbf{x}}.$$

The Euclidean norm of a vector $\|\mathbf{x}\|$ has the following properties:

1. Positivity: $\|\mathbf{x}\| \geq 0$, $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$;
2. Homogeneity: $\|r\mathbf{x}\| = |r|\|\mathbf{x}\|$, $r \in \mathbb{R}$;
3. Triangle Inequality: $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

The Euclidean norm is an example of a general *vector norm*, which is any function satisfying the above three properties of positivity, homogeneity, and triangle inequality. Other examples of vector norms on $\mathbb{R}^n$ include the 1-norm, defined by $\|\mathbf{x}\|_1 = |x_1| + \dots + |x_n|$, and the ∞ -norm, defined by $\|\mathbf{x}\|_\infty = \max_i |x_i|$. The Euclidean norm is often referred to as the 2-norm, and denoted $\|\mathbf{x}\|_2$. The above norms are special cases of the p -norm, given by

$$\|\mathbf{x}\|_p = \begin{cases} (|x_1|^p + \dots + |x_n|^p)^{1/p} & \text{if } 1 \leq p < \infty \\ \max(|x_1|, \dots, |x_n|) & \text{if } p = \infty \end{cases}.$$

Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be an $n \times n$ square matrix. A scalar λ (possibly complex) and a nonzero vector $\mathbf{v}$ satisfying the equation $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ are said to be, respectively, an *eigenvalue* and *eigenvector* of $\mathbf{A}$. For λ to be an eigenvalue it is necessary and sufficient for the matrix $\lambda\mathbf{I} - \mathbf{A}$ to be singular, that is, $\det[\lambda\mathbf{I} - \mathbf{A}] = 0$, where $\mathbf{I}$ is the $n \times n$ identity matrix. This leads to an n th-order polynomial equation

$$\det[\lambda\mathbf{I} - \mathbf{A}] = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = 0.$$

We call the polynomial $\det[\lambda\mathbf{I} - \mathbf{A}]$ the *characteristic polynomial* of the matrix $\mathbf{A}$, and the above equation the *characteristic equation*. According to the fundamental theorem of algebra, the characteristic equation must have n (possibly nondistinct) roots that are the eigenvalues of $\mathbf{A}$.

Concepts from Geometry

Line segments

The *line segment* between two points x and y in $\mathbb{R}^n$ is the set of points on the straight line joining points x and y (see Figure 4.1). Note that if z lies on the line segment between x and y , then

$$z - y = \alpha(x - y),$$

where α is a real number from the interval $[0, 1]$. The above equation can be rewritten as $z = \alpha x + (1 - \alpha)y$. Hence, the line segment between x and y can be represented as

$$\{\alpha x + (1 - \alpha)y : \alpha \in [0, 1]\}.$$

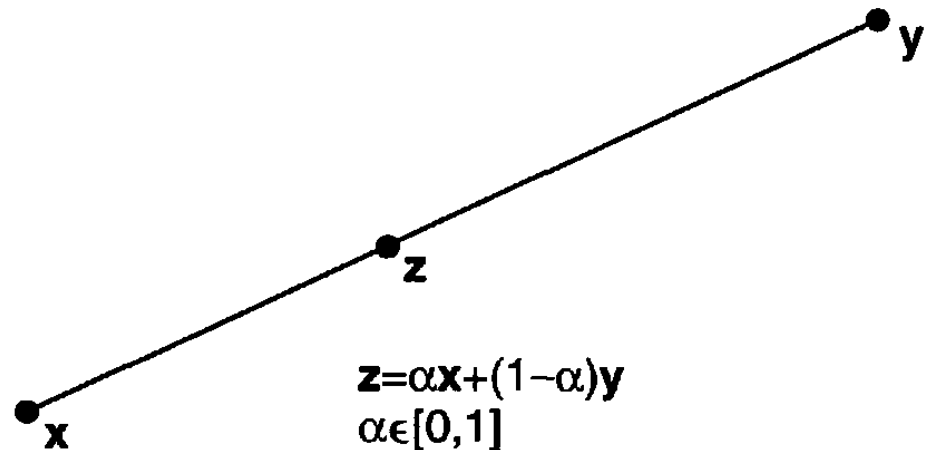


Figure 4.1 A line segment

Hyperplanes

Let $u_1, u_2, \dots, u_n, v \in \mathbb{R}$, where at least one of the u_i is nonzero. The set of all points $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$ that satisfy the linear equation

$$u_1x_1 + u_2x_2 + \cdots + u_nx_n = v$$

is called a *hyperplane* of the space $\mathbb{R}^n$. We may describe the hyperplane by

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{u}^T \mathbf{x} = v\},$$

where

$$\mathbf{u} = [u_1, u_2, \dots, u_n]^T.$$

A hyperplane is not necessarily a subspace of $\mathbb{R}^n$ since, in general, it does not contain the origin. For $n = 2$, the equation of the hyperplane has the form $u_1x_1 + u_2x_2 = v$, which is the equation of a straight line. Thus, straight lines are hyperplanes in $\mathbb{R}^2$. In $\mathbb{R}^3$ (three-dimensional space), hyperplanes are ordinary planes. By translating a hyperplane so that it contains the origin of $\mathbb{R}^n$, it becomes a subspace of $\mathbb{R}^n$ (see Figure 4.2). Because the dimension of this subspace is $n - 1$, we say that the hyperplane has dimension $n - 1$.

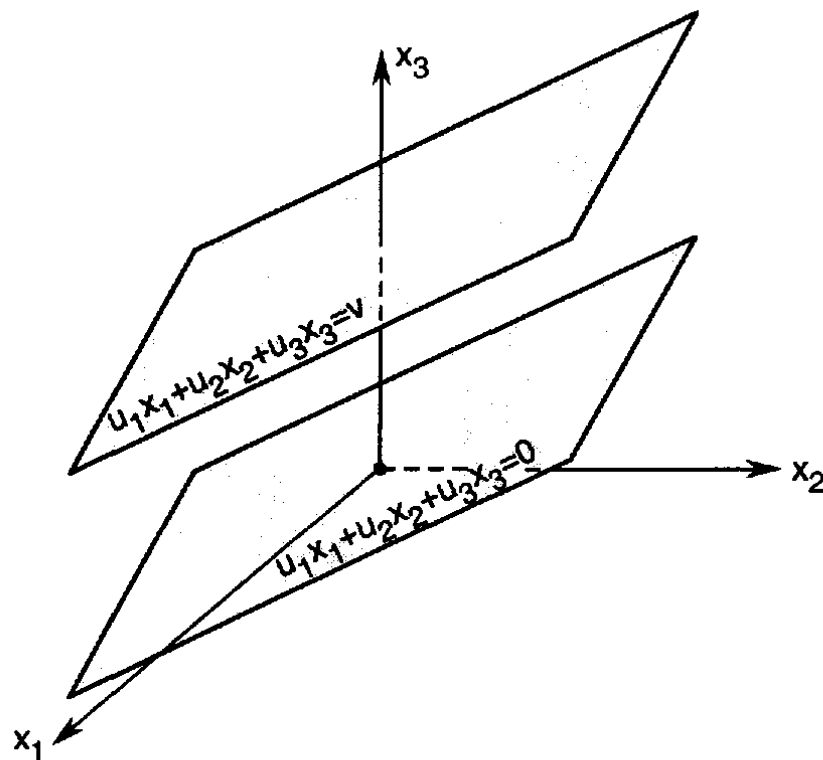


Figure 4.2 Translation of a hyperplane

The hyperplane $H = \{x : u_1x_1 + \cdots + u_nx_n = v\}$ divides $\mathbb{R}^n$ into two *half-spaces*. One of these half-spaces consists of the points satisfying the inequality $u_1x_1 + u_2x_2 + \cdots + u_nx_n \geq v$, denoted

$$H_+ = \{x \in \mathbb{R}^n : u^T x \geq v\},$$

where, as before,

$$\mathbf{u} = [u_1, u_2, \dots, u_n]^T.$$

The other half-space consists of the points satisfying the inequality $u_1x_1 + u_2x_2 + \dots + u_nx_n \leq v$, denoted

$$H_- = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{u}^T \mathbf{x} \leq v\}.$$

The half-space H_+ is called the *positive half-space*, and the half-space H_- is called the *negative half-space*.

Let $\mathbf{a} = [a_1, a_2, \dots, a_n]^T$ be an arbitrary point of the hyperplane H . Thus, $\mathbf{u}^T \mathbf{a} - v = 0$. We can write

$$\begin{aligned} \mathbf{u}^T \mathbf{x} - v &= \mathbf{u}^T \mathbf{x} - v - (\mathbf{u}^T \mathbf{a} - v) \\ &= \mathbf{u}^T (\mathbf{x} - \mathbf{a}) \\ &= u_1(x_1 - a_1) + u_2(x_2 - a_2) + \dots + u_n(x_n - a_n) = 0. \end{aligned}$$

The numbers $(x_i - a_i)$, $i = 1, \dots, n$, are the components of the vector $\mathbf{x} - \mathbf{a}$. Therefore, the hyperplane H consists of the points $\mathbf{x}$ for which $\langle \mathbf{u}, \mathbf{x} - \mathbf{a} \rangle = 0$. In other words, the hyperplane H consists of the points $\mathbf{x}$ for which the vectors $\mathbf{u}$ and $\mathbf{x} - \mathbf{a}$ are orthogonal (see Figure 4.3). We call the vector $\mathbf{u}$ the *normal* to the hyperplane H . The set H_+ consists of those points $\mathbf{x}$ for which $\langle \mathbf{u}, \mathbf{x} - \mathbf{a} \rangle \geq 0$, and H_- consists of those points $\mathbf{x}$ for which $\langle \mathbf{u}, \mathbf{x} - \mathbf{a} \rangle \leq 0$.

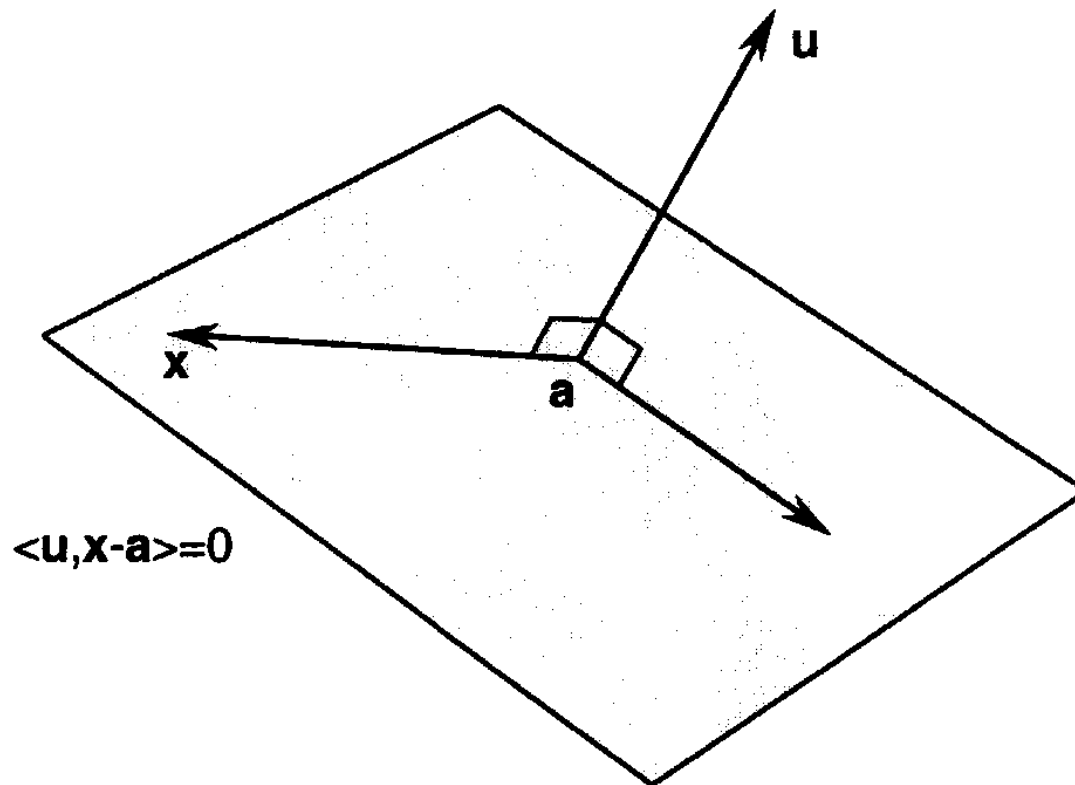


Figure 4.3 The hyperplane $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{u}^T(\mathbf{x} - \mathbf{a}) = 0\}$

Convex Sets

Recall that the line segment between two points $u, v \in \mathbb{R}^n$ is the set $\{w \in \mathbb{R}^n : w = \alpha u + (1 - \alpha)v, \alpha \in [0, 1]\}$. A point $w = \alpha u + (1 - \alpha)v$ (where $\alpha \in [0, 1]$) is called a *convex combination* of the points u and v .

A set $\Theta \subset \mathbb{R}^n$ is *convex* if for all $u, v \in \Theta$, the line segment between u and v is in Θ . Figure 4.4 gives examples of convex sets, whereas Figure 4.5 gives examples of sets that are not convex. Note that Θ is convex if and only if $\alpha u + (1 - \alpha)v \in \Theta$ for all $u, v \in \Theta$ and $\alpha \in (0, 1)$.

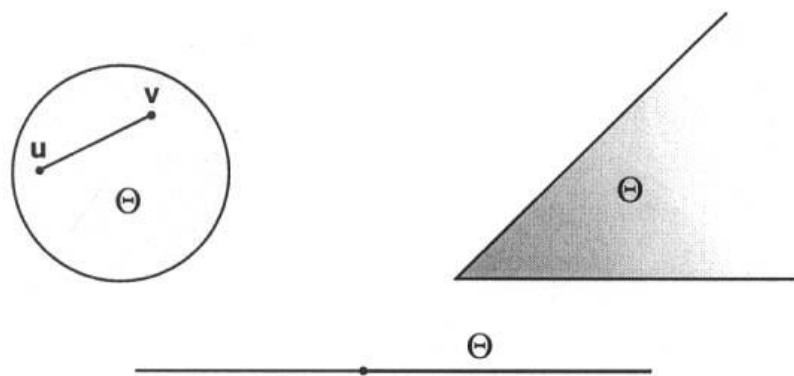


Figure 4.4 Convex sets

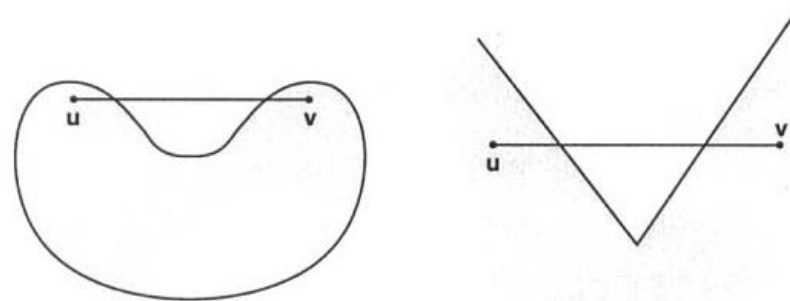


Figure 4.5 Sets that are not convex

Examples of convex sets include:

- the empty set
- a set consisting of a single point
- a line or a line segment
- a subspace
- a hyperplane
- a half-space
- $\mathbb{R}^n$.

A point x in a convex set Θ is said to be an *extreme point* of Θ if there are no two distinct points u and v in Θ such that $x = \alpha u + (1 - \alpha)v$ for some $\alpha \in (0, 1)$. For example, in Figure 4.4, any point on the boundary of the disk is an extreme point, the vertex (corner) of the set on the right is an extreme point, and the endpoint of the half-line is also an extreme point.

Neighborhoods

A *neighborhood* of a point $\mathbf{x} \in \mathbb{R}^n$ is the set

$$\{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{y} - \mathbf{x}\| < \varepsilon\},$$

where ε is some positive number. The neighborhood is also called the *ball* with radius ε and center $\mathbf{x}$.

In the plane $\mathbb{R}^2$, a neighborhood of $\mathbf{x} = [x_1, x_2]^T$ consists of all the points inside of a disc centered at $\mathbf{x}$. In $\mathbb{R}^3$, a neighborhood of $\mathbf{x} = [x_1, x_2, x_3]^T$ consists of all the points inside of a sphere centered at $\mathbf{x}$ (see Figure 4.7).

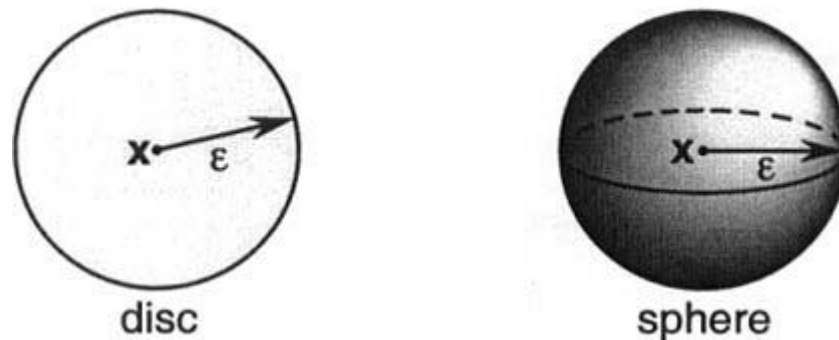


Figure 4.7 Examples of neighborhoods of a point in $\mathbb{R}^2$ and $\mathbb{R}^3$

A point $x \in S$ is said to be an *interior point* of the set S if the set S contains some neighborhood of x , that is, if all points within some neighborhood of x are also in S (see Figure 4.8). The set of all the interior points of S is called the *interior* of S .

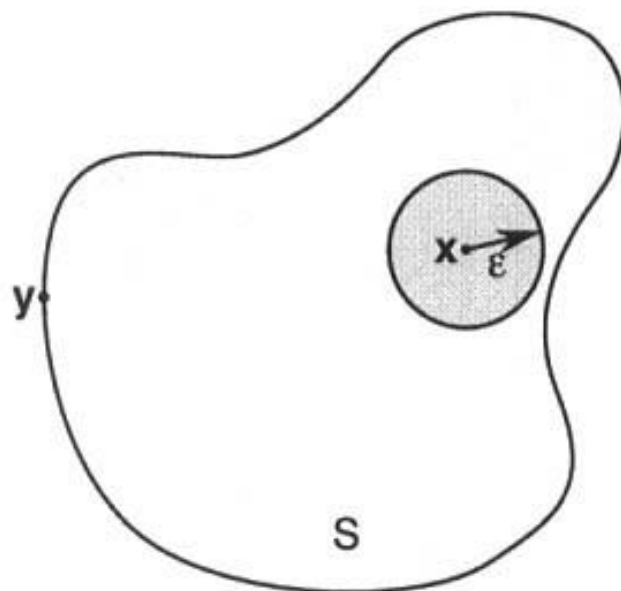


Figure 4.8 x is an interior point, while y is a boundary point

A point x is said to be a *boundary point* of the set S if every neighborhood of x contains a point in S and a point not in S (see Figure 4.8). Note that a boundary point of S may or may not be an element of S . The set of all boundary points of S is called the *boundary* of S .

A set S is said to be *open* if it contains a neighborhood of each of its points, that is, if each of its points is an interior point, or equivalently, if S contains no boundary points.

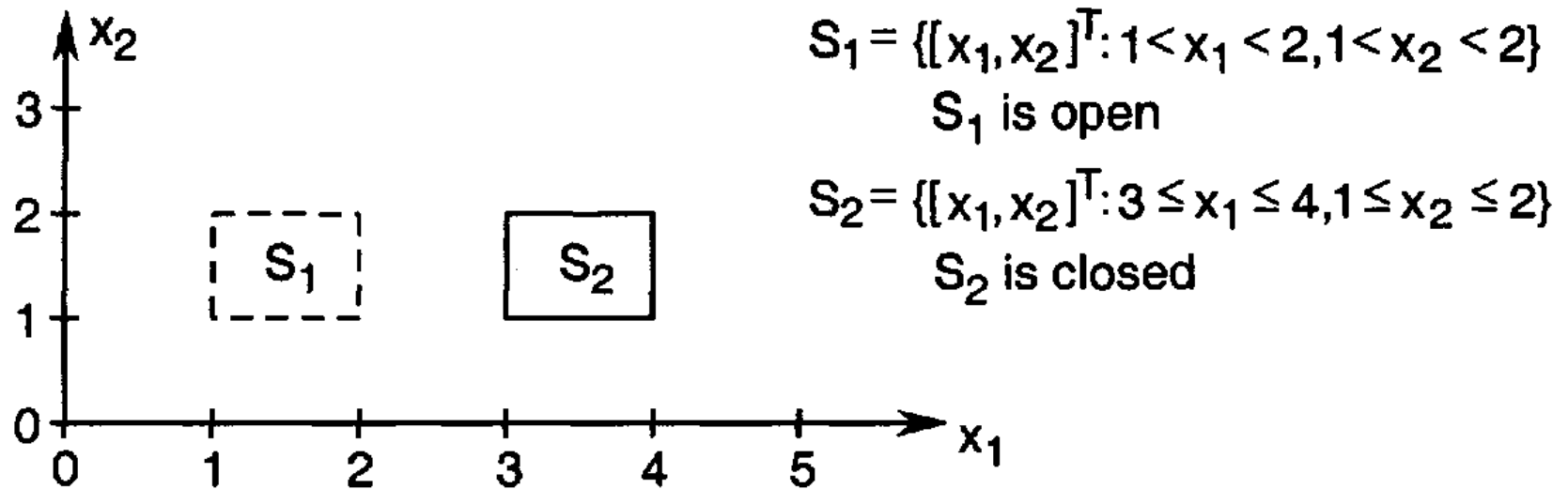


Figure 4.9 Open and closed sets

A set S is said to be *closed* if it contains its boundary (see Figure 4.9). We can show that a set is closed if and only if its complement is open.

A set that is contained in a ball of finite radius is said to be bounded.

A set is compact if it is both closed and bounded.

Optimization Techniques

Lecture 2

Hale Gonce Köçken

**Gradient, Hessian, Convexity of
functions, Definiteness**

Gradient

- When f is real-valued (i.e. $f: \mathbb{R}^n \rightarrow \mathbb{R}$) the derivative $Df(x)$ is a $1 \times n$ matrix, i.e. it is a row vector. Its transpose is called the gradient of the function:

$$\nabla f(x) = Df(x)^T$$

- which is a column vector, i.e. in $\mathbb{R}^n$. Its components are the partial derivatives of f :

$$\nabla f(x)_i = \frac{\partial f(x)}{\partial x_i}, i = 1, 2, \dots, n$$

$$\nabla f(x) = \left[\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right]^T$$

The direction of ∇f is the orientation in which the directional derivative has the largest value and $|\nabla f|$ is the value of that directional derivative.

Hessian Matrix -Second Derivative

Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, if ∇f is differentiable, we say that f is *twice differentiable*, and we write the derivative of ∇f as

$$D^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

The matrix $D^2 f(x)$ is called the *Hessian* matrix of f at x , and is often also denoted $F(x)$.

Find the Hessian Matrix of the function $f(x, y) = x^2y + xy^3$.

We need to first find the first partial derivatives of f . We have that:

$$\frac{\partial f}{\partial x} = 2xy + y^3 \quad , \quad \frac{\partial f}{\partial y} = x^2 + 3xy^2$$

We then calculate the second partial derivatives of f :

$$\frac{\partial^2 f}{\partial x^2} = 2y \quad , \quad \frac{\partial^2 f}{\partial y \partial x} = 2x + 3y^2 \quad , \quad \frac{\partial^2 f}{\partial x \partial y} = 2x + 3y^2 \quad , \quad \frac{\partial^2 f}{\partial y^2} = 6xy$$

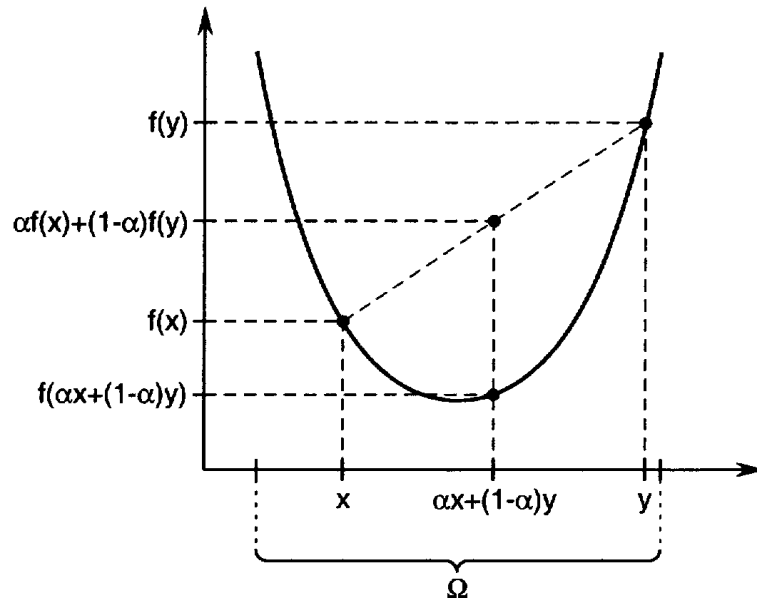
Therefore the Hessian Matrix of f is:

$$\mathcal{H}(x, y) = \begin{bmatrix} 2y & 2x + 3y^2 \\ 2x + 3y^2 & 6xy \end{bmatrix}$$

Example 6.1 Let $f(x_1, x_2) = 5x_1 + 8x_2 + x_1x_2 - x_1^2 - 2x_2^2$.

Theorem 21.2 A function $f : \Omega \rightarrow \mathbb{R}$ defined on a convex set $\Omega \subset \mathbb{R}^n$ is convex if and only if for all $x, y \in \Omega$ and all $\alpha \in (0, 1)$, we have

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y).$$



The line segment between any two points on the graph lies on or above the graph

Figure 21.5 Geometric interpretation of Theorem 21.2

Definition 21.4 A function $f : \Omega \rightarrow \mathbb{R}$ on a convex set $\Omega \subset \mathbb{R}^n$ is *strictly convex* if for all $x, y \in \Omega$, $x \neq y$, and $\alpha \in (0, 1)$, we have

$$f(\alpha x + (1 - \alpha)y) < \alpha f(x) + (1 - \alpha)f(y).$$

Definition 21.5 A function $f : \Omega \rightarrow \mathbb{R}$ on a convex set $\Omega \subset \mathbb{R}^n$ is (strictly) *concave* if $-f$ is (strictly) convex. ■

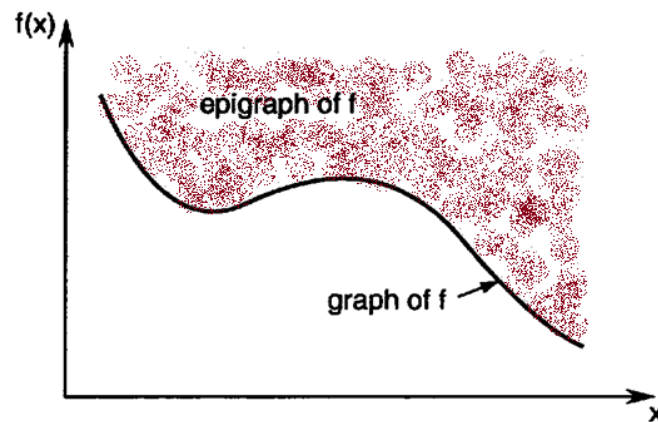
Definition 21.1 The *graph* of $f : \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, is the set of points in $\Omega \times \mathbb{R} \subset \mathbb{R}^{n+1}$ given by

$$\{[x, f(x)]^T : x \in \Omega\}.$$

Definition 21.2 The *epigraph* of a function $f : \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, denoted $\text{epi}(f)$, is the set of points in $\Omega \times \mathbb{R}$ given by

$$\text{epi}(f) = \{[x, \beta]^T : x \in \Omega, \beta \in \mathbb{R}, \beta \geq f(x)\}.$$

The epigraph $\text{epi}(f)$ of a function f is simply the set of points in $\Omega \times \mathbb{R}$ on or above the graph of f (see Figure 21.4). We can also think of $\text{epi}(f)$ as a subset of $\mathbb{R}^{n+1}$.



The hypograph of a function $f: \Omega \rightarrow \mathbb{R}$, denoted $\text{hyp}(f)$, is the set of points in $\Omega \times \mathbb{R}$ given by

$$\text{hyp}(f) = \{[x, \beta]^T: x \in \Omega, \beta \in \mathbb{R}, \beta \leq f(x)\}$$

Theorem (Characterization the convexity of a function in terms of graphs)

- (a) A function f defined on a convex set is concave if and only if its hypograph $\text{hyp } f$ is convex.
- (b) A function f defined on a convex set S is convex if and only if its epigraph $\text{epi } f$ is convex.

Theorem 21.4 *Let $f: \Omega \rightarrow \mathbb{R}$, $f \in C^2$, be defined on an open convex set $\Omega \subset \mathbb{R}^n$. Then, f is convex on Ω if and only if for each $\mathbf{x} \in \Omega$, the Hessian $F(\mathbf{x})$ of f at $\mathbf{x}$ is a positive semidefinite matrix. $\square$*

DEFINITENESS of a MATRIX

- A matrix $H \in R^{n \times n}$ (H is a real symmetric $n \times n$ matrix) is said to be positive definite if for all $h \neq 0$, $h^T H h > 0$.
- The First Method (Definition)

$$\forall h \in R^n, h \neq 0$$

- i) $h^T H h \geq 0 \Rightarrow H$ is positive semidefinite
 $h^T H h > 0 \Rightarrow H$ is positive definite
- ii) $h^T H h \leq 0 \Rightarrow H$ is negative semidefinite
 $h^T H h < 0 \Rightarrow H$ is negative definite
- iii) Otherwise, H is indefinite matrix

The Second Method (Eigenvalues)

- Let the eigenvalues of H be $\lambda_1, \lambda_2, \dots, \lambda_n$.

- i) If all $\lambda_i \geq 0$; $i=1,2,\dots,n$, then H is positive semidefinite
If all $\lambda_i > 0$; $i=1,2,\dots,n$, then H is positive definite
- ii) If all $\lambda_i \leq 0$; $i=1,2,\dots,n$, then H is negative semidefinite
If all $\lambda_i < 0$; $i=1,2,\dots,n$, then H is negative definite
- iii) Otherwise, H is indefinite matrix

The Third Method (Principal Leading Minors)**

A minor of the matrix A of order k is **principal** if it is obtained by deleting $n-k$ rows and the $n-k$ columns with the same numbers.

Note that the definition does not specify which $n-k$ rows and columns to delete, only that their indices must be the same.

The **leading principal minor** of A of order k is the minor of order k obtained by deleting the **last** $n-k$ rows and the **last** $n-k$ columns.

Example 3 For a general 3×3 matrix,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

All principal minors:

There is one third order principal minor, namely $|A|$.

There are three second order principal minors:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \text{ formed by deleting column 3 and row 3;}$$

$$\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, \text{ formed by deleting column 2 and row 2;}$$

$$\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, \text{ formed by deleting column 1 and row 1}$$

And there are three first order principal minors:

$|a_{11}|$, formed by deleting the last two rows and columns

$|a_{22}|$, formed by deleting the first and third rows and columns

$|a_{33}|$, formed by deleting the first two rows and columns

The Leading principal minors:

The first order principal minor:

The second order principal minor:

The third order principal minor:

The algorithm for testing the definiteness of a symmetric matrix:

Let A be a symmetric $n \times n$ matrix, denote

its leading principal minors by Δ_i for $i \in \{1, 2, \dots, n\}$ and

its principal minors by D_i for each order $i \in \{1, 2, \dots, n\}$.

Remark: The matrix A has $\binom{n}{i}$ principal minors of order i .

The algorithm for testing the definiteness of a symmetric matrix:

1. A is **positive definite** if and only if all its n leading principal minors are positive, that is $\Delta_i > 0$ for all $i \in \{1, 2, \dots, n\}$.
2. A is **negative definite** if and only if its n leading principal minors alternate in sign beginning by negative: $\Delta_1 < 0, \Delta_2 > 0, \Delta_3 < 0, \Delta_4 > 0, \dots$
3. A is **positive semidefinite** if and only if all its principal minors are nonnegative, that is all $D_i \geq 0$ for all $i \in \{1, 2, \dots, n\}$ considering all principal minors.
4. A is **negative semidefinite** if and only if every principal minors of odd order is nonpositive (≤ 0) and every principal minors of even order is nonnegative (≥ 0), that is $(-1)^i D_i \geq 0$ for all $i \in \{1, 2, \dots, n\}$ considering all principal minors.
5. Otherwise A is **indefinite**.

Note:

- In the first two cases, it is enough to check the inequality for all the leading principal minors.
- If some i – th order leading principal minor of A is nonzero but does not fit either of the sign patterns in case 1 and case 2, then A is indefinite.
- When some leading principal minor of A is zero, but the others fit one of the patterns in case 1 and case 2: the matrix is not definite, but may or may not be semidefinite. In this case, we must unfortunately check not only the principle leading minors, but **every** principal minor (cases 3 and 4: we must check all principles minors that is for each i with $1 \leq i \leq n$ and for each of the $\binom{n}{i}$ principal minors of order i .)

Example: Determine the definiteness of the following matrix with three methods

$$H = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Example: Determine the definiteness of the following matrices

$$A = \begin{bmatrix} 6 & 4 \\ 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1/2 \end{bmatrix}$$

We have that $D_1 = 6 > 0$, and $D_2 = \begin{vmatrix} 6 & 4 \\ 4 & 5 \end{vmatrix} = 30 - 16 = 14 > 0$.

Therefore, A is a positive definite matrix.

We have that $D_1 = -3 < 0$ and $D_2 = \begin{vmatrix} -3 & 0 \\ 0 & -2 \end{vmatrix} = 6 > 0$. The matrix A is known as a diagonal matrix,

and the determinant $D_3 = \begin{vmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{vmatrix}$

can be computed as the product of the entries in the main diagonal, that is $D_3 = (-3)(-2)(-1) = -6 < 0$.

Since $D_1, D_3 < 0$ and $D_2 > 0$, we have that A is a negative definite matrix.

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & \cancel{1}/2 \end{bmatrix}$$

Theorem 21.4 *Let $f : \Omega \rightarrow \mathbb{R}$, $f \in \mathcal{C}^2$, be defined on an open convex set $\Omega \subset \mathbb{R}^n$. Then, f is convex on Ω if and only if for each $\mathbf{x} \in \Omega$, the Hessian $\mathbf{F}(\mathbf{x})$ of f at $\mathbf{x}$ is a positive semidefinite matrix.* $\square$

Note that by definition of concavity, a function $f : \Omega \rightarrow \mathbb{R}$, $f \in \mathcal{C}^2$, is concave over the convex set $\Omega \subset \mathbb{R}^n$ if and only if for all $\mathbf{x} \in \Omega$, the Hessian $\mathbf{F}(\mathbf{x})$ of f is negative semidefinite.

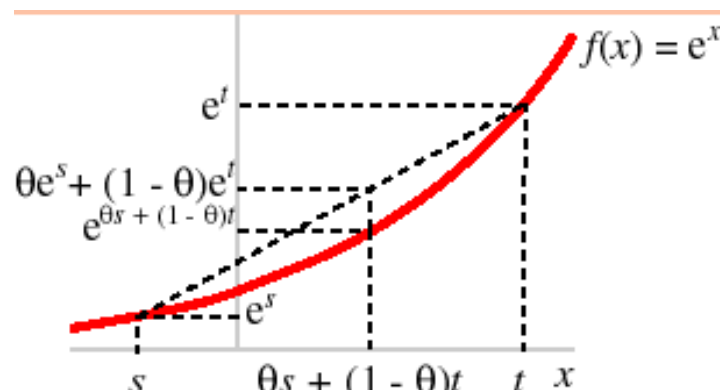
positive semidefinite – convex
positive definite – strictly convex
negative semidefinite – concave
negative definite – strictly concave

Some exercises on convexity of sets and functions

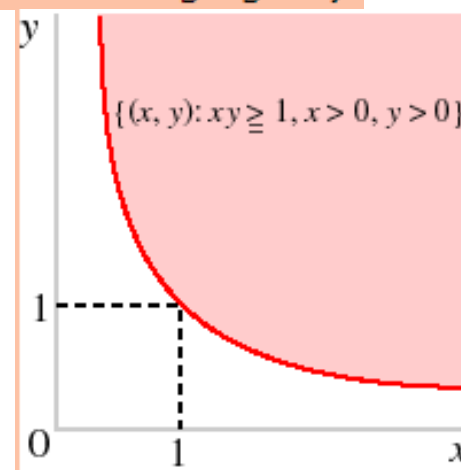
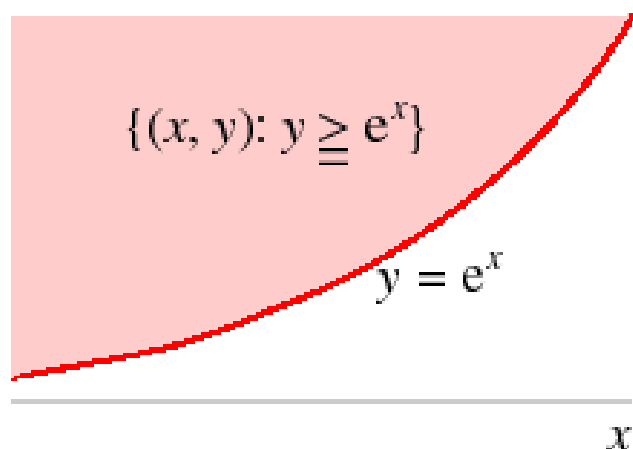
1. By drawing diagrams, determine which of the following sets is convex.

- $\{(x, y): y = e^x\}$.
- $\{(x, y): y \geq e^x\}$.
- $\{(x, y): xy \geq 1, x > 0, y > 0\}$.

a. Not convex, because $e^{\theta x + (1-\theta)u} \neq \theta e^x + (1-\theta)e^u$, as illustrated in the following figure.



b. Convex, because $e^{\theta x + (1-\theta)u} < \theta e^x + (1-\theta)e^u$ (see the following figure).



c. Convex, because if $xy \geq 1$ and $uv \geq 1$ then $(\theta x + (1-\theta)u)(\theta y + (1-\theta)v) \geq 1$ (see figure).

2. For each of the following functions, determine which, if any, of the following conditions the function satisfies: concavity, strict concavity, convexity, strict convexity. (Use whatever technique is most appropriate for each case.)

a. $f(x, y) = x + y$.

b. $f(x, y) = x^2$. [Note: f is a function of two variables.]

c. $f(x, y) = x + y - e^x - e^{x+y}$.

d. $f(x, y, z) = x^2 + y^2 + 3z^2 - xy + 2xz + yz$.

3. Let $f(x_1, x_2) = x_1^2 - x_1x_2 + x_2^2 + 3x_1 - 2x_2 + 1$. Is f convex, concave, or neither?

4. Let $f(x_1, x_2) = x_1^3 + 2x_1^2 + 2x_1x_2 + (1/2)x_2^2 - 8x_1 - 2x_2 - 8$. Find the range of values of (x_1, x_2) for which f is convex, if any.

5. Determine the values of a (if any) for which the function $2x^2 + 2xz + 2ayz + 2z^2$ is concave and the values for which it is convex.

- 6.** Show that the function $-w^2 + 2wx - x^2 - y^2 + 4yz - z^2$ (in the four variables w, x, y , and z) is not concave.

Optimization Techniques

Lecture 3

Hale Gonce Köçken

*Unconstrained Optimization,
Local and Global Minimizer,
Feasible Direction,
FONC and SOSC for a local minimizer*

Unconstrained Optimization,

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{x} \in \Omega. \end{array}$$

The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that we wish to minimize is a real-valued function, and is called the *objective function*, or *cost function*. The vector $\mathbf{x}$ is an n -vector of independent variables, that is, $\mathbf{x} = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$. The variables $x_1, \dots, x_n$ are often referred to as *decision variables*. The set Ω is a subset of $\mathbb{R}^n$, called the *constraint set* or *feasible set*.

The above problem is a general form of a *constrained* optimization problem, because the decision variables are constrained to be in the constraint set Ω . If $\Omega = \mathbb{R}^n$, then we refer to the problem as an *unconstrained* optimization problem.

The constraint " $\mathbf{x} \in \Omega$ " is called a *set constraint*. Often, the constraint set Ω takes the form $\Omega = \{\mathbf{x} : \mathbf{h}(\mathbf{x}) = \mathbf{0}, \mathbf{g}(\mathbf{x}) \leq \mathbf{0}\}$, where $\mathbf{h}$ and $\mathbf{g}$ are given functions. We refer to such constraints as *functional constraints*. The remainder of this chapter deals with general set constraints, including the special case where $\Omega = \mathbb{R}^n$. The case where $\Omega = \mathbb{R}^n$ is called the *unconstrained* case.

minimize	$f(x)$
subject to	$x \in \Omega.$

minimizer of f over Ω
maximizing f is equivalent to minimizing $-f$

Max versus min: Maximizing a function g is equivalent to minimizing $-g$, so there's no loss of generality in concentrating on minimization. This is the convention in much of optimization theory

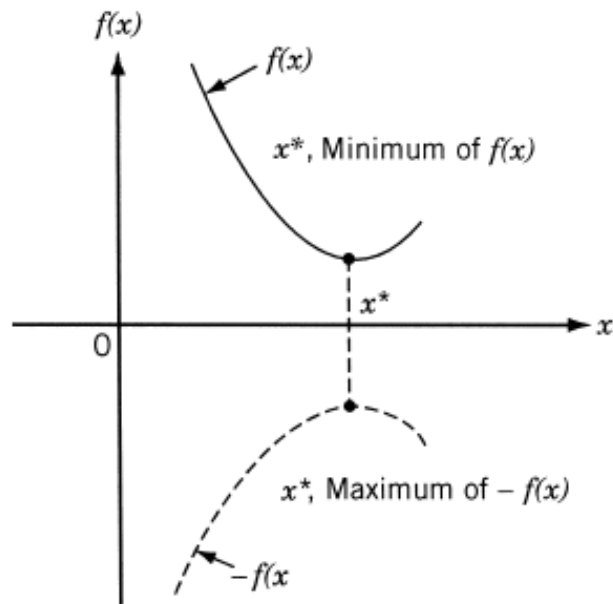


Figure 1.1 Minimum of $f(x)$ is same as maximum of $-f(x)$.

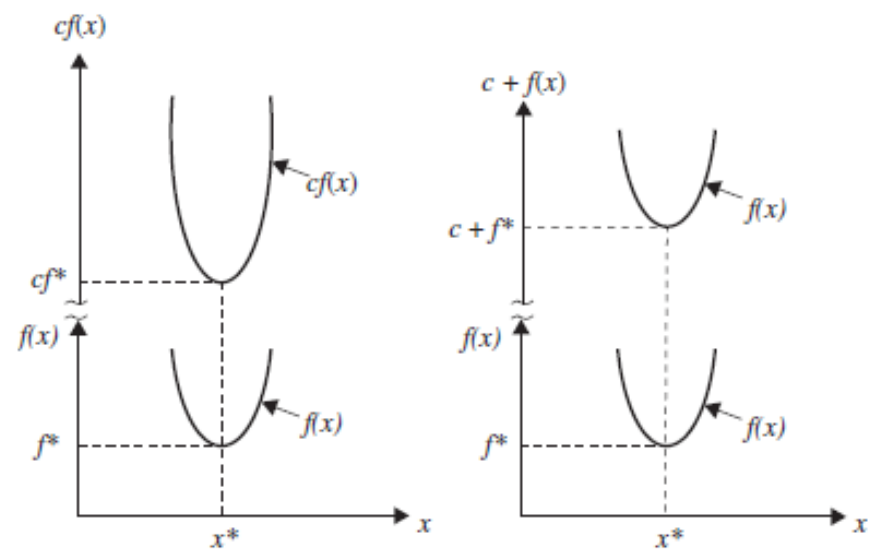


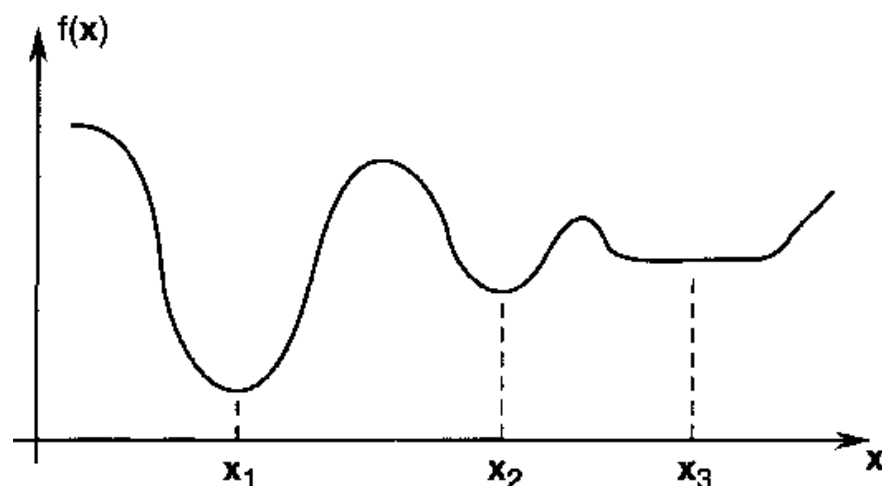
Figure 1.2 Optimum solution of $cf(x)$ or $c + f(x)$ same as that of $f(x)$.

Local and Global Minimizers

Definition 6.1 *Local minimizer.* Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued function defined on some set $\Omega \subset \mathbb{R}^n$. A point $x^* \in \Omega$ is a *local minimizer* of f over Ω if there exists $\varepsilon > 0$ such that $f(x) \geq f(x^*)$ for all $x \in \Omega \setminus \{x^*\}$ and $\|x - x^*\| < \varepsilon$.

Global minimizer. A point $x^* \in \Omega$ is a *global minimizer* of f over Ω if $f(x) \geq f(x^*)$ for all $x \in \Omega \setminus \{x^*\}$.

If, in the above definitions, we replace “ $\geq$ ” with “ $>$ ”, then we have a *strict local minimizer* and a *strict global minimizer*, respectively.



x_1 : strict global minimizer;

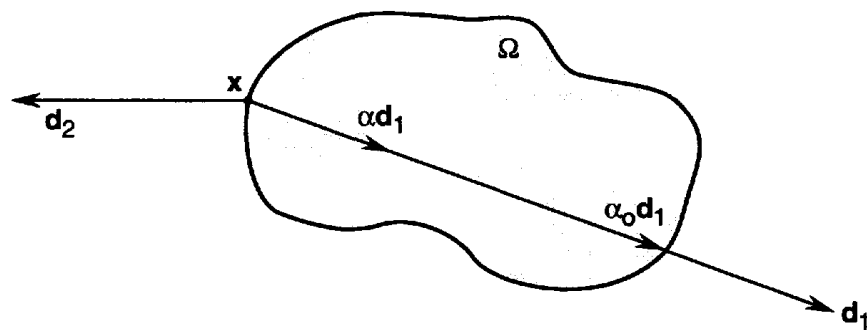
x_2 : strict local minimizer;

x_3 : local (not strict) minimizer

Feasible Direction

Given an optimization problem with constraint set Ω , a minimizer may lie either in the interior or on the boundary of Ω . To study the case where it lies on the boundary, we need the notion of *feasible directions*.

Definition 6.2 Feasible direction. A vector $\mathbf{d} \in \mathbb{R}^n$, $\mathbf{d} \neq \mathbf{0}$, is a *feasible direction* at $\mathbf{x} \in \Omega$ if there exists $\alpha_0 > 0$ such that $\mathbf{x} + \alpha\mathbf{d} \in \Omega$ for all $\alpha \in [0, \alpha_0]$.



$\mathbf{d}_1$ is a feasible direction,

$\mathbf{d}_2$ is not a feasible direction

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function and let $\mathbf{d}$ be a feasible direction at $\mathbf{x} \in \Omega$. The *directional derivative of f in the direction $\mathbf{d}$* , denoted $\partial f / \partial \mathbf{d}$, is the real-valued function defined by

$$\frac{\partial f}{\partial \mathbf{d}}(\mathbf{x}) = \lim_{\alpha \rightarrow 0} \frac{f(\mathbf{x} + \alpha\mathbf{d}) - f(\mathbf{x})}{\alpha}.$$

If $\|\mathbf{d}\| = 1$, then $\partial f / \partial \mathbf{d}$ is the rate of increase of f at $\mathbf{x}$ in the direction $\mathbf{d}$. To compute the above directional derivative, suppose that $\mathbf{x}$ and $\mathbf{d}$ are given. Then, $f(\mathbf{x} + \alpha \mathbf{d})$ is a function of α , and

$$f(\mathbf{x} + \alpha \mathbf{d}) = F(\alpha)$$

$$\begin{aligned} F'(\alpha) &= \frac{\delta f}{\delta x_1} \frac{dx_1}{d\alpha} + \frac{\delta f}{\delta x_2} \frac{dx_2}{d\alpha} + \cdots + \frac{\delta f}{\delta x_n} \frac{dx_n}{d\alpha} \\ &= \nabla f(\mathbf{x})^T \cdot \mathbf{d} \end{aligned}$$

In summary, if $\mathbf{d}$ is a unit vector, that is, $\|\mathbf{d}\| = 1$, then $\langle \nabla f(\mathbf{x}), \mathbf{d} \rangle$ is the rate of increase of f at the point $\mathbf{x}$ in the direction $\mathbf{d}$.

Theorem ***First-Order Necessary Condition (FONC).** Let Ω be a subset of $\mathbb{R}^n$ and $f \in C^1$ a real-valued function on Ω . If $\mathbf{x}^*$ is a local minimizer of f over Ω , and $\mathbf{x}^*$ is an interior point of Ω , then*

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

What happens if Ω is an open subset of $\mathbb{R}^n$?

FONC: *Let Ω be an open set and f is a continuously differentiable function over Ω ($f \in C^1$).*

If $\mathbf{x}^ \in \Omega$ is a minimizer of f , then $\nabla f(\mathbf{x}^*) = \mathbf{0}$.*

$$\nabla f(x^*) = 0.$$

This is the **first-order necessary condition for optimality**. The condition is "first-order" because it is derived using the first-order expansion. We emphasize that the result is valid when $f \in C^1$ and x^* is an interior point of Ω .

Definition: A point x^* satisfying the FONC condition $\nabla f(x^*) = 0$ is called a ***stationary point***.

Definition: A function f has **critical points** at all points x_0 where $f'(x_0) = 0$ or $f(x)$ is not differentiable.

In several variables case, the same definition can be given as:

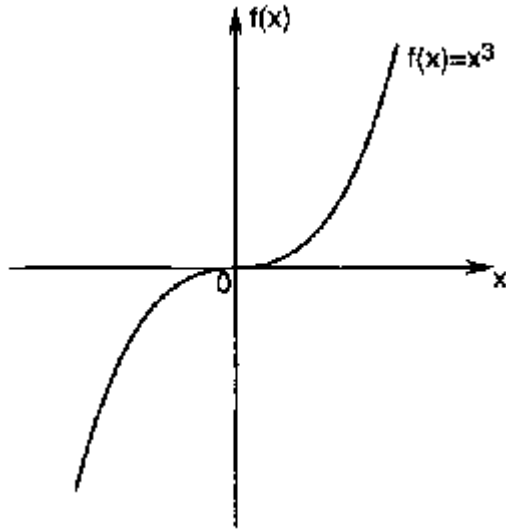
A function with several variables has **critical points** where the gradient is 0 or any of the partial derivatives is not defined.

Second Order Sufficient Condition (SOSC):

Let Ω be an open set and f is a twice continuously differentiable function over Ω ($f \in C^2$). The sufficient condition for a local minimizer ($x^* \in \Omega$) is that the Hessian matrix at x^* is positive semidefinite.

Proof of SOSC:

Example 6.5 Consider a function of one variable $f(x) = x^3$, $f : \mathbb{R} \rightarrow \mathbb{R}$



The point 0 satisfies the FONC but is not a minimizer

This point is an inflection point.

Definition: An inflection point is a point on a curve at which the concavity changes.

A necessary condition for x to be an inflection point is $f''(x) = 0$. A sufficient condition requires $f''(x + \varepsilon)$ and $f''(x - \varepsilon)$ to have opposite signs in the neighborhood of x .

In multivariable calculus, a point of a function or surface which is a stationary point but not an extremum is called a saddle point.

In summary

FONC: Solve the following equation and obtain all the stationary points.

$$\nabla f(x^*) = 0.$$

SOSC: Form the Hessian at each of the stationary points:

If the Hessian is *positive* definite, then that point is a *strict* local minimum,

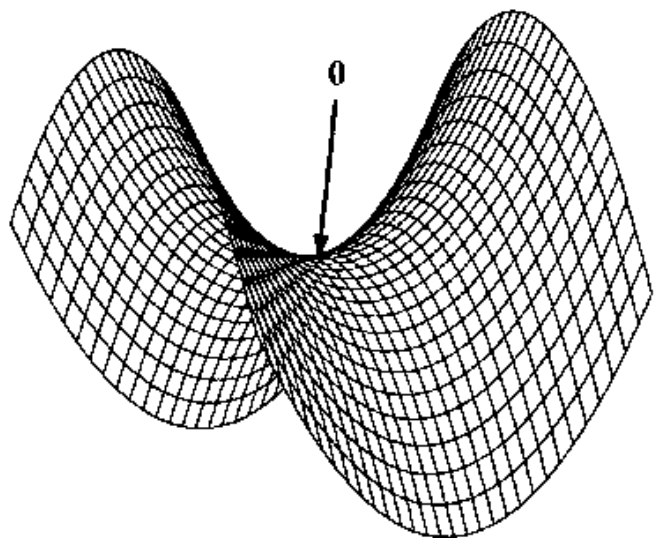
If the Hessian is *positive* semidefinite, then that point is a local minimum,

If the Hessian is *negative* definite, then that point is a *strict* local maximum,

If the Hessian is *negative* semidefinite, then that point is a local maximum.

If the Hessian is indefinite, then that point is a saddle point.

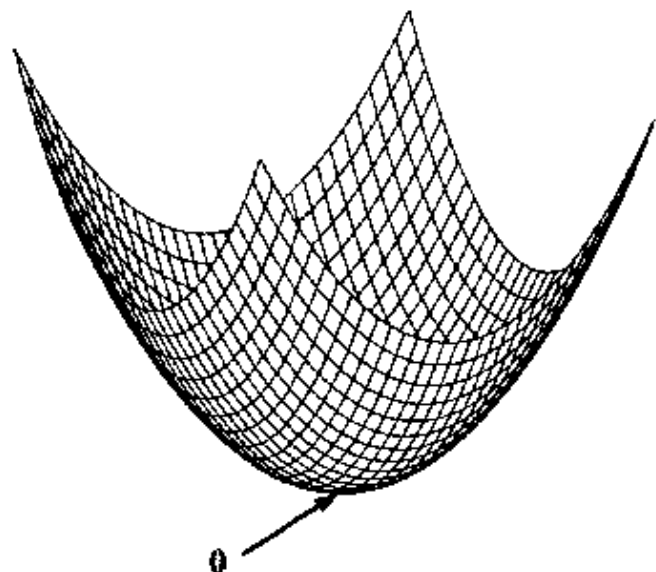
Example 6.6 Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, where $f(x) = x_1^2 - x_2^2$.



The point $\mathbf{0}$ satisfies the FONC but not SOSC;

this point is not a minimizer

Example 6.7 Let $f(x) = x_1^2 + x_2^2$.



$x = [0, 0]^T$ is a strict global minimizer.

Example: A small startup company produces speakers and subwoofers for computers that they sell through a website. After extensive research, the company has developed a revenue function,

$$R(x, y) = x(110 - 4.5x) + y(155 - 2y) \text{ thousand dollars}$$

where x is the number of subwoofers produced and sold in thousands and y is the number of speakers produced and sold in thousands. The corresponding cost function is

$$C(x, y) = 3x^2 + 3y^2 + 5xy - 5y + 50 \text{ thousand dollars}$$

Find the production levels that maximize revenue.

Find and classify all the critical points of the following functions:

1) $f(x_1, x_2) = x^3 - x^2y + 2y^2.$

$$2)f(x, y) = (y - 2)x^2 - y^2$$

$$3) f(x, y) = 7x - 8y + 2xy - x^2 + y^3$$

$$4) f = x_1^3 + x_2^3 + x_3^3 - 3x_1^2x_2 - 3x_2^2x_3 - 3x_3^2x_1 + 6x_1 + 6x_2 + 6x_3$$

$$f(x, y) = x + \frac{1}{2x + y} - \frac{1}{x + y}$$

$$\nabla f = \begin{pmatrix} 1 - \frac{2}{(2x+y)^2} + \frac{1}{(x+y)^2} \\ -\frac{1}{(2x+y)^2} + \frac{1}{(x+y)^2} \end{pmatrix} \quad H(f) = \begin{pmatrix} \frac{8}{(2x+y)^3} - \frac{2}{(x+y)^3}; & \frac{4}{(2x+y)^3} - \frac{2}{(x+y)^3} \\ \frac{4}{(2x+y)^3} - \frac{2}{(x+y)^3}; & \frac{2}{(2x+y)^3} - \frac{2}{(x+y)^3} \end{pmatrix}$$

Need to solve the equations $\nabla f(x, y) = 0$:

$$1 - \frac{2}{(2x + y)^2} + \frac{1}{(x + y)^2} = 0, \quad -\frac{1}{(2x + y)^2} + \frac{1}{(x + y)^2} = 0$$

Subtracting the second equation from the first one we see that $(2x + y)^2 = 1$ and $(x + y)^2 = 1$ which give us 4 critical points

$$1) (0, 1); \quad 2) (0, -1); \quad 3) (2, -3); \quad 4) (-2, 3).$$

$$1) H(f)|_{(0,1)} = \begin{pmatrix} 6 & 2 \\ 2 & 0 \end{pmatrix},$$

$$2) H(f)|_{(0,-1)} = \begin{pmatrix} -6 & -2 \\ -2 & 0 \end{pmatrix}$$

$$3) H(f)|_{(2,-3)} = \begin{pmatrix} 10 & 6 \\ 6 & 4 \end{pmatrix},$$

$$4) H(f)|_{(-2,3)} = \begin{pmatrix} -10 & -6 \\ -6 & -4 \end{pmatrix},$$

Practical Example

A container with an open top is to have 10 m^3 capacity and be made of thin sheet metal. Calculate the dimensions of the box if it is to use the minimum possible amount of metal.

Solution: Let A be the total area of metal used to make the box, and let x and y be the length and width and z the height. Then

$$A = 2xz + 2yz + xy$$

Also

$$xyz = 10$$

because the volume is 10 m^3 . This implies that $z = \frac{10}{xy}$. Putting this into the formula for A gives A as a function of x and y only:

$$\begin{aligned} A &= 2x \left(\frac{10}{xy} \right) + 2y \left(\frac{10}{xy} \right) + xy \\ &= \frac{20}{y} + \frac{20}{x} + xy \end{aligned}$$

We shall apply our techniques to this function. Now

$$\frac{\partial A}{\partial x} = -\frac{20}{x^2} + y, \quad \frac{\partial A}{\partial y} = -\frac{20}{y^2} + x$$

and for a stationary point we need $\partial A/\partial x = \partial A/\partial y = 0$. this gives

$$y = \frac{20}{x^2} \quad \text{and} \quad x = \frac{20}{y^2}.$$

Therefore

$$y = \frac{20}{(20/y^2)^2} = \frac{y^4}{20}$$

Since the zero root $y = 0$ is obviously not consistent with having a volume of 10 m^3 we reject $y = 0$ and conclude that $y^3 = 20$ so that $y = 20^{1/3} = 2.714$ metres. From $x = 20/y^2$ we conclude $x = 2.714$ metres also. To find z , use $z = \frac{10}{xy}$ so that $z = 1.357 \text{ m}$.

We have to show that these values do indeed give a **minimum**. Now

$$\frac{\partial^2 A}{\partial x \partial y} = 1, \quad \frac{\partial^2 A}{\partial x^2} = \frac{40}{x^3}, \quad \frac{\partial^2 A}{\partial y^2} = \frac{40}{y^3}$$

So, when $(x, y) = (2.714, 2.714)$,

$$A_{xx}A_{yy} - A_{xy}^2 = (2)(2) - 1^2 = 3 > 0$$

so it is either a max or a min. But $A_{xx} > 0$ and $A_{yy} > 0$ so it is a minimum. Our conclusion is that the box should have length 2.714 m, width 2.714 m and height 1.357 m. The actual area of metal used will then (from the formula for A) be 22.1 m^2 .

Optimization Techniques

Lecture 4

Hale Gonce Köçken

Iterative methods for unconstrained or set-constrained optimization

Gradient methods

The method of steepest descent

The method of steepest descent for a quadratic function

Iterative methods for unconstrained or set-constrained optimization

- In our last lecture, we have seen a theoretical basis for the solution of nonlinear unconstrained problems.
- Suppose that one is confronted with a highly nonlinear function of 20 variables.
- Then the FONC requires the solution of 20 nonlinear simultaneous equations for 20 variables.
- These equations, being nonlinear, will normally have multiple solutions. In addition, we would have to compute 210 second derivatives (provided f in C^2) to use the SOSC.
- Thus, we will now concern with iterative methods of solving such problems.

Gradients Methods

Let $\mathbf{x}^{(0)}$ be a starting point, and consider the point $\mathbf{x}^{(0)} - \alpha \nabla f(\mathbf{x}^{(0)})$.

Then, by Taylor's theorem we obtain

$$f(\mathbf{x}^{(0)} - \alpha \nabla f(\mathbf{x}^{(0)})) = f(\mathbf{x}^{(0)}) - \alpha \|\nabla f(\mathbf{x}^{(0)})\|^2 + o(\alpha^2).$$

if $\nabla f(\mathbf{x}^{(0)}) \neq \mathbf{0}$, then for sufficiently small $\alpha > 0$, we have

$$f(\mathbf{x}^{(0)} - \alpha \nabla f(\mathbf{x}^{(0)})) < f(\mathbf{x}^{(0)}).$$

This means that the point $\mathbf{x}^{(0)} - \alpha \nabla f(\mathbf{x}^{(0)})$ is an improvement over the point $\mathbf{x}^{(0)}$ if we are searching for a minimizer.

To formulate an algorithm that implements the above idea, suppose that we are given a point $\mathbf{x}^{(k)}$. To find the next point $\mathbf{x}^{(k+1)}$, we start at $\mathbf{x}^{(k)}$ and move by an amount $-\alpha_k \nabla f(\mathbf{x}^{(k)})$, where α_k is a positive scalar called the *step size*. The above procedure leads to the following iterative algorithm:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha_k \nabla f(\mathbf{x}^{(k)}).$$

Remember that the function f increases more in the direction of the gradient than in any other direction.

A numerical example

The gradient vector of a scalar function $f(x_1, x_2, \dots, x_n)$ is defined as a column vector

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix}^T = \mathbf{c}$$

For example

$$f(x_1, x_2) = 25x_1^2 + x_2^2 \quad \Longrightarrow \quad \mathbf{c} = \nabla f = \begin{bmatrix} 2(25)x_1^* \\ 2x_2^* \end{bmatrix} = \begin{bmatrix} 2(25)(.6) \\ 2(4) \end{bmatrix} = \begin{bmatrix} 30 \\ 8 \end{bmatrix}$$

at the point $x_1^* = .6, x_2^* = 4$

The normalized gradient vector

$$\bar{\mathbf{c}} = \frac{\mathbf{c}}{\sqrt{\mathbf{c}^T \mathbf{c}}}$$

For example, at the point $x_1^* = .6, x_2^* = 4$

$$\bar{\mathbf{c}} = \frac{1}{\sqrt{30^2 + 8^2}} \begin{bmatrix} 30 \\ 8 \end{bmatrix} = \begin{bmatrix} .96625 \\ .2577 \end{bmatrix}$$

The gradient vector represents a direction of maximum rate of increase for

the function $f(\mathbf{x})$ at $\mathbf{x}^*$. For example,

$f(.6, 4) = 25(.6)^2 + 4^2 = 25$ If we increase $\mathbf{x}$ in the direction $\bar{\mathbf{c}}$ by a step size of $\alpha = .5$

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha \bar{\mathbf{c}} = \begin{bmatrix} .6 \\ 4 \end{bmatrix} + .5 \begin{bmatrix} .96625 \\ .2577 \end{bmatrix} = \begin{bmatrix} 1.083125 \\ 4.12885 \end{bmatrix}$$

The function value becomes $f(\mathbf{x}^{(1)}) = 25(1.083125)^2 + (4.122885)^2 = 46.327$

If we move in a direction $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha \bar{\mathbf{c}} = \begin{bmatrix} .6 \\ 4 \end{bmatrix} + .5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1.1 \\ 4 \end{bmatrix}$$

The function value becomes

$$f(\mathbf{x}^{(1)}) = 25(1.1)^2 + (4)^2 = 46.25$$

If we move in a direction $\begin{bmatrix} 0 & 1 \end{bmatrix}^T$

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha \bar{\mathbf{c}} = \begin{bmatrix} .6 \\ 4 \end{bmatrix} + .5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} .6 \\ 4.5 \end{bmatrix}$$

The function value becomes

$$f(\mathbf{x}^{(1)}) = 25(.6)^2 + (4.5)^2 = 29.25$$

We can see that moving along the gradient direction results in the maximum increase in the function.

The maximum rate of change of $f(\mathbf{x})$ at any point $\mathbf{x}^*$ is the magnitude of the gradient vector given by $\|\mathbf{c}\| = \sqrt{\mathbf{c}^T \mathbf{c}}$

HW: Find the maximum rate of change of $f(x, y) = xe^{-y} + 3y$ at the point $(1, 0)$ and the direction in which it occurs.

Solution: The maximum rate of change occurs in the direction of the gradient vector $\nabla f(x, y)$, and the maximum rate of change is at $|\nabla f(x, y)|$.

$$\nabla f(x, y) = \langle e^{-y}, -xe^{-y} + 3 \rangle$$

$$\nabla f(1, 0) = \langle 1, 2 \rangle$$

$$|\nabla f(1, 0)| = |\langle 1, 2 \rangle| = \sqrt{5}$$

Thus the maximum rate of change is $\sqrt{5}$ in the direction $\nabla f(1, 0) = [1 \ 2]^T$.

Thus, the direction in which $\nabla f(\mathbf{x})$ points is the direction of maximum rate of increase of f at $\mathbf{x}$. The direction in which $-\nabla f(\mathbf{x})$ points is the direction of maximum rate of decrease of f at $\mathbf{x}$. Hence, the direction of negative gradient is a good direction to search if we want to find a function minimizer.

The method of steepest descent

The method of steepest descent is a gradient algorithm where the step size α_k is chosen to achieve the maximum amount of decrease of the objective function at each individual step. Specifically, α_k is chosen to minimize $\phi_k(\alpha) \triangleq f(x^{(k)} - \alpha \nabla f(x^{(k)}))$. In other words,

$$\alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} - \alpha \nabla f(x^{(k)})).$$

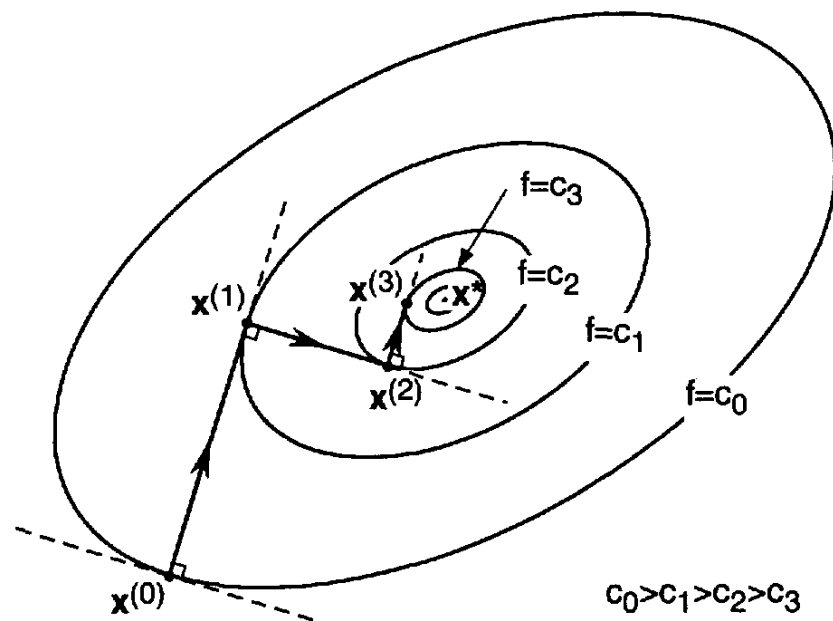


Figure 8.2 Typical sequence resulting from the method of steepest descent

In summary, the steepest descent algorithm can be given as follows:

Step 1: Choose $x^{(0)}$.

Step 2: Calculate $\nabla f(x^{(i)})$. If $\nabla f(x^{(i)}) = 0$, stop.

Step 3: Determine the next point $x^{(i+1)}$ with

$x^{(i+1)} = x^{(i)} - \alpha_i \nabla f(x^{(i)})$ where α_i is chosen to minimize the function $f(x^{(i)} - \alpha_i \nabla f(x^{(i)}))$. Set $i = i + 1$ and go to Step 2.

Example : Let $f(x, y) = x^2 + y^2$. Find a minimizer of f with the method of steepest descent assuming the initial point as $(1,1)$.

The method of steepest descent for a quadratic function

Let us now see what the method of steepest descent does with a quadratic function of the form

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{b}^T \mathbf{x},$$

where $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is a symmetric positive definite matrix, $\mathbf{b} \in \mathbb{R}^n$, and $\mathbf{x} \in \mathbb{R}^n$. The unique minimizer of f can be found by setting the gradient of f to zero, where

$$\nabla f(\mathbf{x}) = \mathbf{Q} \mathbf{x} - \mathbf{b},$$

because $D(\mathbf{x}^T \mathbf{Q} \mathbf{x}) = \mathbf{x}^T (\mathbf{Q} + \mathbf{Q}^T) = 2\mathbf{x}^T \mathbf{Q}$, and $D(\mathbf{b}^T \mathbf{x}) = \mathbf{b}^T$.

The Hessian of f is $\mathbf{F}(\mathbf{x}) = \mathbf{Q} = \mathbf{Q}^T > 0$.

To simplify the notation we write $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$.

Then, the steepest descent algorithm for the quadratic function can be represented as

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha_k \mathbf{g}^{(k)},$$

$$\begin{aligned} \text{where } \alpha_k &= \arg \min_{\alpha \geq 0} f(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}) \\ &= \arg \min_{\alpha \geq 0} \left(\frac{1}{2} (\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)})^T \mathbf{Q} (\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}) - (\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)})^T \mathbf{b} \right). \end{aligned}$$

In the quadratic case, we can find an explicit formula for α_k . We proceed as follows. Assume $\mathbf{g}^{(k)} \neq \mathbf{0}$, for if $\mathbf{g}^{(k)} = \mathbf{0}$, then $\mathbf{x}^{(k)} = \mathbf{x}^*$ and the algorithm stops. Because $\alpha_k \geq 0$ is a minimizer of $\phi_k(\alpha) = f(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)})$, we apply the FONC to $\phi_k(\alpha)$ to obtain

$$\phi'_k(\alpha) = (\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)})^T \mathbf{Q} (-\mathbf{g}^{(k)}) - \mathbf{b}^T (-\mathbf{g}^{(k)}).$$

Therefore, $\phi'_k(\alpha) = 0$ if $\alpha \mathbf{g}^{(k)T} \mathbf{Q} \mathbf{g}^{(k)} = (\mathbf{x}^{(k)T} \mathbf{Q} - \mathbf{b}^T) \mathbf{g}^{(k)}$. But

$$\mathbf{x}^{(k)T} \mathbf{Q} - \mathbf{b}^T = \mathbf{g}^{(k)T}.$$

Hence,

$$\alpha_k = \frac{\mathbf{g}^{(k)T} \mathbf{g}^{(k)}}{\mathbf{g}^{(k)T} \mathbf{Q} \mathbf{g}^{(k)}}.$$

In summary, the method of steepest descent for the quadratic takes the form

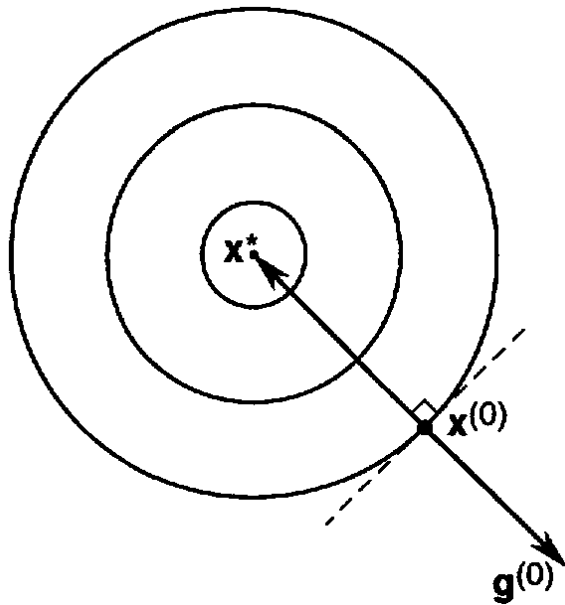
$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \left(\frac{\mathbf{g}^{(k)T} \mathbf{g}^{(k)}}{\mathbf{g}^{(k)T} \mathbf{Q} \mathbf{g}^{(k)}} \right) \mathbf{g}^{(k)},$$

where

$$\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)}) = \mathbf{Q} \mathbf{x}^{(k)} - \mathbf{b}.$$

Example 8.2 Let

$$f(x_1, x_2) = x_1^2 + x_2^2.$$



Then, starting from an arbitrary initial point $x^{(0)} \in \mathbb{R}^2$ we arrive at the solution $x^* = \mathbf{0} \in \mathbb{R}^2$ in only one step. See Figure 8.6.

However, if

$$f(x_1, x_2) = \frac{x_1^2}{5} + x_2^2,$$

then the method of steepest descent shuffles ineffectively back and forth when searching for the minimizer in a narrow valley (see Figure 8.7). This example illustrates a major drawback in the steepest descent method. More sophisticated methods that alleviate this problem are discussed in subsequent chapters.

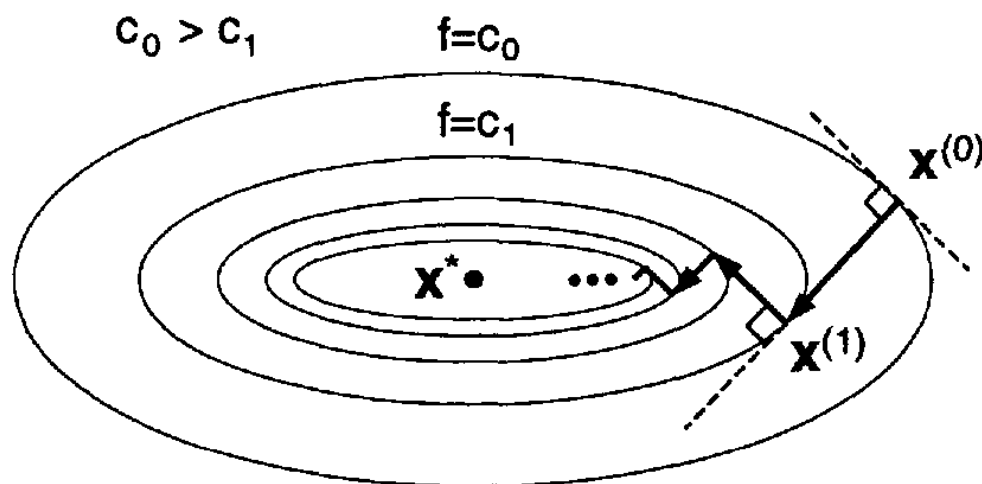


Figure 8.7 Steepest descent method in search for minimizer in a narrow valley

The condition $\nabla f(\mathbf{x}^{(k+1)}) = \mathbf{0}$, however, is not directly suitable as a practical stopping criterion, because the numerical computation of the gradient will rarely be identically equal to zero.

A practical stopping criterion is to check if the norm $\|\nabla f(\mathbf{x}^{(k)})\|$ of the gradient is less than a prespecified threshold, in which case we stop.

Alternatively, we may compute the absolute difference $|f(\mathbf{x}^{(k+1)}) - f(\mathbf{x}^{(k)})|$ between objective function values for every two successive iterations, and if the difference is less than some prespecified threshold, then we stop; that is, we stop when

$$|f(\mathbf{x}^{(k+1)}) - f(\mathbf{x}^{(k)})| < \varepsilon,$$

where $\varepsilon > 0$ is a prespecified threshold.

Yet another alternative is to compute the norm $\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|$ of the difference between two successive iterates, and we stop if the norm is less than a prespecified threshold:

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\| < \varepsilon.$$

Alternatively, we may check “relative” values of the above quantities; for example,

$$\frac{|f(\mathbf{x}^{(k+1)}) - f(\mathbf{x}^{(k)})|}{|f(\mathbf{x}^{(k)})|} < \varepsilon,$$

or

$$\frac{\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|}{\|\mathbf{x}^{(k)}\|} < \varepsilon.$$

Note that the above stopping criteria are relevant to all the iterative algorithms we discuss in this part.

With the stopping criteria modification, the steepest descent algorithm can be given as follows:

Step 1: Choose $x^{(0)}$.

Step 2: Calculate $\nabla f(x^{(i)})$. If $\nabla f(x^{(i)}) = 0$ or the selected stopping criteria is satisfied, then stop.

Step 3: Determine the next point $x^{(i+1)}$ with $x^{(i+1)} = x^{(i)} - \alpha_i \nabla f(x^{(i)})$ where α_i is chosen to minimize the function $f(x^{(i)} - \alpha_i \nabla f(x^{(i)}))$. Set $i = i + 1$ and go to Step 2.

Example : Let $f(x, y) = x^2 + y^2 + xy - 3x$. Find a minimizer of f with

a) Analytical method

b) The method of steepest descent with the following stopping criteria

i. $\|\nabla f(x)\| < 0.8,$

ii. $|f(x^{(k+1)}) - f(x^{(k)})| < 0.2.$

Point	f value	$\mathbf{g}_k$	α_k
$x^{(0)} = (0,0)^T$	0	$\begin{bmatrix} -3 \\ 0 \end{bmatrix}$	0.5
$x^{(1)} = (3/2, 0)^T$	-2.25	$\begin{bmatrix} 0 \\ 3/2 \end{bmatrix}$	0.5
$x^{(2)} = (3/2, -3/4)^T$	-2.8116	$\begin{bmatrix} -0.75 \\ 0 \end{bmatrix}$	0.5
$x^{(3)} = (1.875, -0.75)^T$	-2.9531	$\begin{bmatrix} 0 \\ 0.375 \end{bmatrix}$	0.5
$x^{(4)} = (1.875, -0.9375)^T$	-2.9883	$\begin{bmatrix} -0.1875 \\ 0 \end{bmatrix}$	0.5
$x^{(5)} = (1.9688, -0.9375)^T$	-2.9971	$\begin{bmatrix} 0.001 \\ 0.0938 \end{bmatrix}$	0.4995
$x^{(6)} = (1.9688, -0.9844)^T$	-2.9993	$\begin{bmatrix} -0.0468 \\ 0 \end{bmatrix}$	0.5
$x^{(7)} = (1.9922, -0.9844)^T$	-2.9998	$\begin{bmatrix} 0 \\ 0.0234 \end{bmatrix}$	0.5

Point	f value	$\mathbf{g}_k$	α_k
$x^{(8)} = (1.9922, -0.9961)^T$	-3	$\begin{bmatrix} -0.0117 \\ 0 \end{bmatrix}$	0.5
$x^{(9)} = (1.9981, -0.9961)^T$	-3	$\begin{bmatrix} 0.001 \\ 0.0059 \end{bmatrix}$	0.4917
$x^{(10)} = (1.9981, -0.9990)^T$	-3	$\begin{bmatrix} -0.0028 \\ 0.0001 \end{bmatrix}$	0.5185
$x^{(11)} = (1.9996, -0.9991)^T$	-3	$\begin{bmatrix} 0.001 \\ 0.0014 \end{bmatrix}$	0.4668
$x^{(12)} = (1.9996, -0.9998)^T$	-3	$\begin{bmatrix} 0.001 \\ -0.6 \end{bmatrix}$	0.5
$x^{(13)} = (1.9999, -0.9998)^T$	-3	$\begin{bmatrix} 0.001 \\ 0.0003 \end{bmatrix}$	0.5
$x^{(14)} = (1.9999, -1)^T$	-3	$\begin{bmatrix} -0.0002 \\ -0.0001 \end{bmatrix}$	0.3571
$x^{(15)} = (2, -1)^T$	-3	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	

Quadratic Function

$\theta \in R^{n \times n}$ is a positive definite matrix and $x \in R^n$;

$$f(x) = \frac{1}{2} x' \theta x - x' b$$

For $n=2$, $\theta \in R^{2 \times 2}$, $x \in R^2$

$$f(x) = \frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} ax + cy & bx + dy \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - [ex + fy]$$

$$= \frac{1}{2} [ax^2 + cxy + bxy + dy^2] - [ex + fy]$$

$$f(x) = \frac{1}{2} ax^2 + \frac{1}{2} dy^2 + \frac{1}{2} (b+c)xy - ex + fy$$

$$\theta = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

a : twice the coefficient of x^2

d : twice the coefficient of y^2

$b+c$: twice the coefficient of xy

e : reverse sign of the coefficient of x

f : reverse sign of the coefficient of y

The derivative of a quadratic function

$$\left. \begin{aligned} \frac{\partial f}{\partial x} &= ax + \frac{(b+c)}{2}y - e \\ \frac{\partial f}{\partial y} &= dy + \frac{(b+c)}{2}x - f \end{aligned} \right\} \Rightarrow \nabla f(x) = \begin{bmatrix} a & \frac{(b+c)}{2} \\ \frac{(b+c)}{2} & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} e \\ f \end{bmatrix}$$

Since θ is a symmetrical matrix, then $b=c$. Thus, we have

$$\nabla f(x) = \begin{bmatrix} a & b \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} e \\ f \end{bmatrix} = \theta x - b \quad \text{and also} \quad \nabla^2 f = \begin{bmatrix} a & b \\ b & d \end{bmatrix} = \theta.$$

Optimization Techniques

Lecture 5

Hale Gonce Köçken

The Newton's method

Newton's Method

Recall that the method of steepest descent uses only first derivatives (gradients) in selecting a suitable search direction. This strategy is not always the most effective. If higher derivatives are used, the resulting iterative algorithm may perform better than the steepest descent method.

Newton's method (sometimes called the Newton-Raphson method) uses first and second derivatives and indeed does perform better than the steepest descent method if the initial point is close to the minimizer.

idea behind this method is as follows. Given a starting point, we construct a quadratic approximation to the objective function that matches the first and second derivative values at that point.

We then minimize the approximate (quadratic) function instead of the original objective function. We use the minimizer of the approximate function as the starting point in the next step and repeat the procedure iteratively.

If the objective function is quadratic, then the approximation is exact, and the method yields the true minimizer in one step.

If, on the other hand, the objective function is not quadratic, then the approximation will provide only an estimate of the position of the true minimizer.

Figure 9.1 illustrates the above idea.

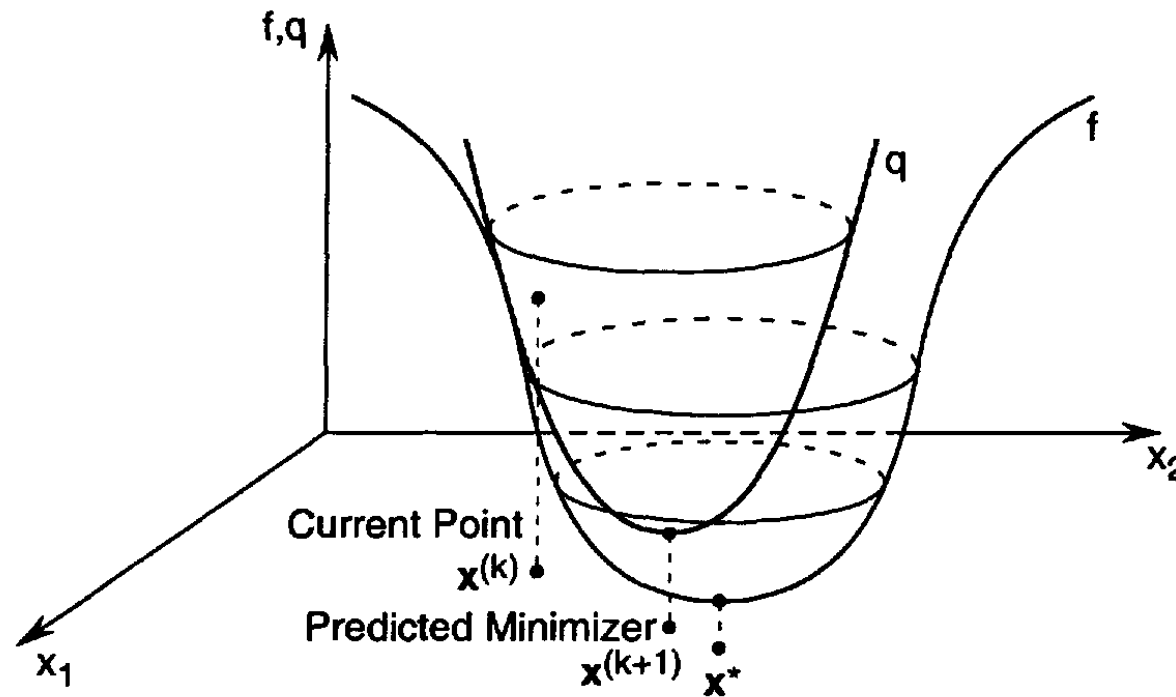


Figure 9.1 Quadratic approximation to the objective function using first and second derivatives

We can obtain a quadratic approximation to the given twice continuously differentiable objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ using the Taylor series expansion of f about the current point $\mathbf{x}^{(k)}$, neglecting terms of order three and higher. We obtain

$$f(\mathbf{x}) \approx f(\mathbf{x}^{(k)}) + (\mathbf{x} - \mathbf{x}^{(k)})^T \mathbf{g}^{(k)} + \frac{1}{2}(\mathbf{x} - \mathbf{x}^{(k)})^T \mathbf{F}(\mathbf{x}^{(k)})(\mathbf{x} - \mathbf{x}^{(k)}) \triangleq q(\mathbf{x}),$$

where, for simplicity, we use the notation $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$. Applying the FONC to q yields

$$\mathbf{0} = \nabla q(\mathbf{x}) = \mathbf{g}^{(k)} + \mathbf{F}(\mathbf{x}^{(k)})(\mathbf{x} - \mathbf{x}^{(k)}).$$

If $\mathbf{F}(\mathbf{x}^{(k)}) > 0$, then q achieves a minimum at

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \mathbf{F}(\mathbf{x}^{(k)})^{-1} \mathbf{g}^{(k)}.$$

This recursive formula represents Newton's method.

Example: Find the minimizer of $f(x) = 2x_1^2 + 2x_1x_2 + 2x_2^2 - 4x_1 - 6x_2$ with the starting point

$x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ by using Newton's method.

Optimization Techniques

Lecture 6

Hale Gonce Köçken

The conjugate direction methods

Conjugate Direction Methods

The class of *conjugate direction methods* can be viewed as being intermediate between the method of steepest descent and Newton's method. The conjugate direction methods have the following properties:

1. Solve quadratics of n variables in n steps;
2. The usual implementation, the *conjugate gradient algorithm*, requires no Hessian matrix evaluations;
3. No matrix inversion and no storage of an $n \times n$ matrix required.

Basically, two directions $\mathbf{d}^{(1)}$ and $\mathbf{d}^{(2)}$ in $\mathbb{R}^n$ are said to be Q -conjugate if $\mathbf{d}^{(1)T} Q \mathbf{d}^{(2)} = 0$.

Definition 10.1 Let Q be a real symmetric $n \times n$ matrix. The directions $\mathbf{d}^{(0)}, \mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \dots, \mathbf{d}^{(m)}$ are Q -conjugate if, for all $i \neq j$, we have $\mathbf{d}^{(i)T} Q \mathbf{d}^{(j)} = 0$.

Lemma 10.1 *Let $\mathbf{Q}$ be a symmetric positive definite $n \times n$ matrix. If the directions $\mathbf{d}^{(0)}, \mathbf{d}^{(1)}, \dots, \mathbf{d}^{(k)} \in \mathbb{R}^n$, $k \leq n - 1$, are nonzero and $\mathbf{Q}$ -conjugate, then they are linearly independent.*

Proof. Let $\alpha_0, \dots, \alpha_k$ be scalars such that

$$\alpha_0 \mathbf{d}^{(0)} + \alpha_1 \mathbf{d}^{(1)} + \dots + \alpha_k \mathbf{d}^{(k)} = \mathbf{0}.$$

Premultiplying the above equality by $\mathbf{d}^{(j)T} \mathbf{Q}$, $0 \leq j \leq k$, yields

$$\alpha_j \mathbf{d}^{(j)T} \mathbf{Q} \mathbf{d}^{(j)} = 0,$$

because all other terms $\mathbf{d}^{(j)T} \mathbf{Q} \mathbf{d}^{(i)} = 0$, $i \neq j$, by $\mathbf{Q}$ -conjugacy. But $\mathbf{Q} = \mathbf{Q}^T > 0$ and $\mathbf{d}^{(j)} \neq \mathbf{0}$; hence $\alpha_j = 0$, $j = 0, 1, \dots, k$. Therefore, $\mathbf{d}^{(0)}, \mathbf{d}^{(1)}, \dots, \mathbf{d}^{(k)}$, $k \leq n - 1$, are linearly independent.

Example 10.1 Let

$$\mathbf{Q} = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

Note that $\mathbf{Q} = \mathbf{Q}^T > 0$. The matrix $\mathbf{Q}$ is positive definite because all its leading principal minors are positive:

Our goal is to construct a set of $\mathbf{Q}$ -conjugate vectors $\mathbf{d}^{(0)}, \mathbf{d}^{(1)}, \mathbf{d}^{(2)}$.

Let $\mathbf{d}^{(0)} = [1, 0, 0]^T$, $\mathbf{d}^{(1)} = [d_1^{(1)}, d_2^{(1)}, d_3^{(1)}]^T$, $\mathbf{d}^{(2)} = [d_1^{(2)}, d_2^{(2)}, d_3^{(2)}]^T$. We require $\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(1)} = 0$. We have

$$\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(1)} = [1, 0, 0] \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} d_1^{(1)} \\ d_2^{(1)} \\ d_3^{(1)} \end{bmatrix}$$

Let $d_1^{(1)} = 1, d_2^{(1)} = 0, d_3^{(1)} = -3$. Then, $\mathbf{d}^{(1)} = [1, 0, -3]^T$, and thus $\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(1)} = 0$.

To find the third vector $\mathbf{d}^{(2)}$, which would be $\mathbf{Q}$ -conjugate with $\mathbf{d}^{(0)}$ and $\mathbf{d}^{(1)}$, we require $\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(2)} = 0$ and $\mathbf{d}^{(1)T} \mathbf{Q} \mathbf{d}^{(2)} = 0$. We have

$$\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(2)} = 3d_1^{(2)} + d_3^{(2)} = 0,$$

$$\mathbf{d}^{(1)T} \mathbf{Q} \mathbf{d}^{(2)} = -6d_2^{(2)} - 8d_3^{(2)} = 0.$$

If we take $\mathbf{d}^{(2)} = [1, 4, -3]^T$, then the resulting set of vectors is mutually conjugate.

A systematic procedure for finding $\mathbf{Q}$ -conjugate vectors

$$\mathbf{Q} \mathbf{x} = \mathbf{b}.$$

1. Basic Conjugate Direction Algorithm

We now present the conjugate direction algorithm for minimizing the quadratic function of n variables

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{x}^T \mathbf{b},$$

where $\mathbf{Q} = \mathbf{Q}^T > 0$, $\mathbf{x} \in \mathbb{R}^n$. Note that because $\mathbf{Q} > 0$,

Given a starting point $\mathbf{x}^{(0)}$, and $\mathbf{Q}$ -conjugate directions $\mathbf{d}^{(0)}, \mathbf{d}^{(1)}, \dots, \mathbf{d}^{(n-1)}$; for $k \geq 0$,

$$\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)}) = \mathbf{Q} \mathbf{x}^{(k)} - \mathbf{b},$$

$$\alpha_k = -\frac{\mathbf{g}^{(k)T} \mathbf{d}^{(k)}}{\mathbf{d}^{(k)T} \mathbf{Q} \mathbf{d}^{(k)}},$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)}.$$

Theorem 10.1 *For any starting point $\mathbf{x}^{(0)}$, the basic conjugate direction algorithm converges to the unique $\mathbf{x}^*$ (that solves $\mathbf{Q} \mathbf{x} = \mathbf{b}$) in n steps; that is, $\mathbf{x}^{(n)} = \mathbf{x}^*$.*

Example 10.2 Find the minimizer of

$$f(x_1, x_2) = \frac{1}{2} \mathbf{x}^T \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \mathbf{x} - \mathbf{x}^T \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \mathbf{x} \in \mathbb{R}^2,$$

using the conjugate direction method with the initial point $\mathbf{x}^{(0)} = [0, 0]^T$, and Q -conjugate directions $\mathbf{d}^{(0)} = [1, 0]^T$ and $\mathbf{d}^{(1)} = [-\frac{3}{8}, \frac{3}{4}]^T$.

We have $\mathbf{g}^{(0)} = -\mathbf{b} = [1, -1]^T$,

and hence
$$\alpha_0 = -\frac{\mathbf{g}^{(0)T} \mathbf{d}^{(0)}}{\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(0)}} =$$

Thus, $\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha_0 \mathbf{d}^{(0)} =$

To find $\mathbf{x}^{(2)}$, we compute $\mathbf{g}^{(1)} = \mathbf{Q}\mathbf{x}^{(1)} - \mathbf{b} =$

$$\text{and } \alpha_1 = -\frac{\mathbf{g}^{(1)T}\mathbf{d}^{(1)}}{\mathbf{d}^{(1)T}\mathbf{Q}\mathbf{d}^{(1)}} =$$

$$\text{Therefore, } \mathbf{x}^{(2)} = \mathbf{x}^{(1)} + \alpha_1\mathbf{d}^{(1)} =$$

Because f is a quadratic function in two variables, $\mathbf{x}^{(2)} = \mathbf{x}^*$.

2. Conjugate Gradient Algorithm

Our first search direction from an initial point $\mathbf{x}^{(0)}$ is in the direction of steepest descent; that is,

$$\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}.$$

Thus, $\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha_0 \mathbf{d}^{(0)}$, where

$$\alpha_0 = \arg \min_{\alpha \geq 0} f(\mathbf{x}^{(0)} + \alpha \mathbf{d}^{(0)}) = -\frac{\mathbf{g}^{(0)T} \mathbf{d}^{(0)}}{\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(0)}}.$$

In the next stage, we search in a direction $\mathbf{d}^{(1)}$ that is $\mathbf{Q}$ -conjugate to $\mathbf{d}^{(0)}$. We choose $\mathbf{d}^{(1)}$ as a linear combination of $\mathbf{g}^{(1)}$ and $\mathbf{d}^{(0)}$. In general, at the $(k+1)$ st step, we choose $\mathbf{d}^{(k+1)}$ to be a linear combination of $\mathbf{g}^{(k+1)}$ and $\mathbf{d}^{(k)}$. Specifically, we choose

$$\mathbf{d}^{(k+1)} = -\mathbf{g}^{(k+1)} + \beta_k \mathbf{d}^{(k)}, \quad k = 0, 1, 2, \dots$$

The coefficients $\beta_k, k = 1, 2, \dots$, are chosen in such a way that $\mathbf{d}^{(k+1)}$ is $\mathbf{Q}$ -conjugate to $\mathbf{d}^{(0)}, \mathbf{d}^{(1)}, \dots, \mathbf{d}^{(k)}$. This is accomplished by choosing β_k to be

$$\beta_k = \frac{\mathbf{g}^{(k+1)T} \mathbf{Q} \mathbf{d}^{(k)}}{\mathbf{d}^{(k)T} \mathbf{Q} \mathbf{d}^{(k)}}.$$

The conjugate gradient algorithm is summarized below.

1. Set $k := 0$; select the initial point $\mathbf{x}^{(0)}$.
2. $\mathbf{g}^{(0)} = \nabla f(\mathbf{x}^{(0)})$. If $\mathbf{g}^{(0)} = \mathbf{0}$, stop, else set $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$.
3. $\alpha_k = -\frac{\mathbf{g}^{(k)T} \mathbf{d}^{(k)}}{\mathbf{d}^{(k)T} \mathbf{Q} \mathbf{d}^{(k)}}.$
4. $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)}.$
5. $\mathbf{g}^{(k+1)} = \nabla f(\mathbf{x}^{(k+1)})$. If $\mathbf{g}^{(k+1)} = \mathbf{0}$, stop.
6. $\beta_k = \frac{\mathbf{g}^{(k+1)T} \mathbf{Q} \mathbf{d}^{(k)}}{\mathbf{d}^{(k)T} \mathbf{Q} \mathbf{d}^{(k)}}.$
7. $\mathbf{d}^{(k+1)} = -\mathbf{g}^{(k+1)} + \beta_k \mathbf{d}^{(k)}.$
8. Set $k := k + 1$; go to step 3.

Example 10.3 Consider the quadratic function

$$f(x_1, x_2, x_3) = \frac{3}{2}x_1^2 + 2x_2^2 + \frac{3}{2}x_3^2 + x_1x_3 + 2x_2x_3 - 3x_1 - x_3.$$

We find the minimizer using the conjugate gradient algorithm, using the starting point $\mathbf{x}^{(0)} = [0, 0, 0]^T$.

We can represent f as $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} - \mathbf{x}^T \mathbf{b}$,

$$\text{where } \mathbf{Q} = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}.$$

We have

$$\mathbf{g}(\mathbf{x}) = \nabla f(\mathbf{x}) = \mathbf{Q}\mathbf{x} - \mathbf{b} = [3x_1 + x_3 - 3, 4x_2 + 2x_3, x_1 + 2x_2 + 3x_3 - 1]^T.$$

$$\text{Hence, } \mathbf{g}^{(0)} = [-3, 0, -1]^T, \quad \mathbf{d}^{(0)} = -\mathbf{g}^{(0)},$$

$$\alpha_0 = -\frac{\mathbf{g}^{(0)T} \mathbf{d}^{(0)}}{\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(0)}} = \frac{10}{36} = 0.2778,$$

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha_0 \mathbf{d}^{(0)} = [0.8333, 0, 0.2778]^T.$$

The next stage yields

$$\mathbf{g}^{(1)} = \nabla f(\mathbf{x}^{(1)}) = [-0.2222, 0.5556, 0.6667]^T,$$

$$\beta_0 = \frac{\mathbf{g}^{(1)T} \mathbf{Q} \mathbf{d}^{(0)}}{\mathbf{d}^{(0)T} \mathbf{Q} \mathbf{d}^{(0)}} = 0.08025.$$

We can now compute

$$\mathbf{d}^{(1)} = -\mathbf{g}^{(1)} + \beta_0 \mathbf{d}^{(0)} = [0.4630, -0.5556, -0.5864]^T.$$

Hence,

$$\alpha_1 = -\frac{\mathbf{g}^{(1)T} \mathbf{d}^{(1)}}{\mathbf{d}^{(1)T} \mathbf{Q} \mathbf{d}^{(1)}} = 0.2187,$$

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} + \alpha_1 \mathbf{d}^{(1)} = [0.9346, -0.1215, 0.1495]^T.$$

To perform the third iteration, we compute

$$\begin{aligned} \mathbf{g}^{(2)} &= \nabla f(\mathbf{x}^{(2)}) = [-0.04673, -0.1869, 0.1402]^T, \\ \beta_1 &= \frac{\mathbf{g}^{(2)T} \mathbf{Q} \mathbf{d}^{(1)}}{\mathbf{d}^{(1)T} \mathbf{Q} \mathbf{d}^{(1)}} = 0.07075, \\ \mathbf{d}^{(2)} &= -\mathbf{g}^{(2)} + \beta_1 \mathbf{d}^{(1)} = [0.07948, 0.1476, -0.1817]^T. \end{aligned}$$

Hence,

$$\alpha_2 = -\frac{\mathbf{g}^{(2)T} \mathbf{d}^{(2)}}{\mathbf{d}^{(2)T} \mathbf{Q} \mathbf{d}^{(2)}} = 0.8231,$$

and

$$\mathbf{x}^{(3)} = \mathbf{x}^{(2)} + \alpha_2 \mathbf{d}^{(2)} = [1.000, 0.000, 0.000]^T.$$

Note that

$$\mathbf{g}^{(3)} = \nabla f(\mathbf{x}^{(3)}) = \mathbf{0},$$

as expected, because f is a quadratic function of three variables. Hence, $\mathbf{x}^* = \mathbf{x}^{(3)}$.

3. Conjugate Gradient Algorithm for non-quadratic problems

For a quadratic, the matrix Q , the Hessian of the quadratic, is constant.

However, for a general nonlinear function the Hessian is a matrix that has to be reevaluated at each iteration of the algorithm.

This can be computationally very expensive. Thus, an efficient implementation of the conjugate gradient algorithm that eliminates the Hessian evaluation at each step is desirable.

The Hestenes-Stiefel formula.

Recall that $\beta_k = \frac{\mathbf{g}^{(k+1)T} \mathbf{Q} \mathbf{d}^{(k)}}{\mathbf{d}^{(k)T} \mathbf{Q} \mathbf{d}^{(k)}}.$

$$\mathbf{Q} \mathbf{d}^{(k)} \longleftrightarrow (\mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}) / \alpha_k.$$

The two terms are equal in the quadratic case, as we now show. Now, $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)}$. Premultiplying both sides by $\mathbf{Q}$, and recognizing that $\mathbf{g}^{(k)} = \mathbf{Q} \mathbf{x}^{(k)} - \mathbf{b}$, we get $\mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha_k \mathbf{Q} \mathbf{d}^{(k)}$, which we can rewrite as $\mathbf{Q} \mathbf{d}^{(k)} = (\mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}) / \alpha_k$. Substituting this into the original equation for β_k gives

$$\beta_k = \frac{\mathbf{g}^{(k+1)T} [\mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}]}{\mathbf{d}^{(k)T} [\mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}]},$$

which is called the Hestenes-Stiefel formula.

The Polak-Ribière formula.

$$\beta_k = \frac{\mathbf{g}^{(k+1)T} [\mathbf{g}^{(k+1)} - \mathbf{g}^{(k)}]}{\mathbf{g}^{(k)T} \mathbf{g}^{(k)}}.$$

The Fletcher-Reeves formula.

$$\beta_k = \frac{\mathbf{g}^{(k+1)T} \mathbf{g}^{(k+1)}}{\mathbf{g}^{(k)T} \mathbf{g}^{(k)}},$$

For nonquadratic problems, the algorithm will not usually converge in n steps, and as the algorithm progresses, the “ Q -conjugacy” of the direction vectors will tend to deteriorate. Thus, a common practice is to reinitialize the direction vector to the negative gradient after every few iterations (e.g., n or $n + 1$), and continue until the algorithm satisfies the stopping criterion.

Example:

Let $f(\mathbf{x})$, $\mathbf{x} = [x_1, x_2]^T \in \mathbb{R}^2$, be given by

$$f(\mathbf{x}) = \frac{5}{2}x_1^2 + \frac{1}{2}x_2^2 + 2x_1x_2 - 3x_1 - x_2.$$

- a. Express $f(\mathbf{x})$ in the form of $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{x}^T \mathbf{b}$.
- b. Find the minimizer of f using the conjugate gradient algorithm. Use a starting point of $\mathbf{x}^{(0)} = [0, 0]^T$.
- c. Calculate the minimizer of f analytically from $\mathbf{Q}$ and $\mathbf{b}$, and check it with your answer in part b.

Nonlinear Constrained Optimization

1. *Problems with equality constraints*
2. *Problems with equality and inequality constraints*
3. *Problems under the nonnegativity constraints*

In this part, we discuss methods for solving a class of nonlinear constrained optimization problems that can be formulated as:

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) \\ & \text{subject to} && h_i(\mathbf{x}) = 0, && i = 1, \dots, m \\ & && g_j(\mathbf{x}) \leq 0, && j = 1, \dots, p, \end{aligned}$$

where $\mathbf{x} \in \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$, and $m \leq n$. In vector notation, the problem above can be represented in the following *standard form*:

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) \\ & \text{subject to} && \mathbf{h}(\mathbf{x}) = \mathbf{0} \\ & && \mathbf{g}(\mathbf{x}) \leq \mathbf{0}, \end{aligned}$$

where $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^p$. As usual, we adopt the following terminology.

Definition 19.1 Any point satisfying the constraints is called a *feasible point*. The set of all feasible points

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{h}(\mathbf{x}) = \mathbf{0}, \mathbf{g}(\mathbf{x}) \leq \mathbf{0}\}$$

is called the *feasible set*. ■

As we remarked in Part II, there is no loss of generality by considering only minimization problems. For if we are confronted with a maximization problem, it can be easily transformed into the minimization problem by observing that

$$\text{maximize } f(\mathbf{x}) = \text{minimize } -f(\mathbf{x}).$$

1. Problems with equality constraint – Lagrange Multipliers

Consider the following problem.

Min $f(x)$

s.t. $g_i(x)=0 \quad i=1,2,\dots,k \leq n$

where $f(x)$ and $g_i(x)$ are continuously differentiable functions in $\mathbb{R}^n$.

In mathematical optimization, the method of lagrange multipliers is a strategy for finding local maxima and minima of a function subject to equality constraints.

For the above problem, we introduce new variables λ_i ($i=1,2,\dots,k$) called Lagrange multipliers and study the Lagrange function (or Lagrangian) defined by

If x^* is a minimizer of f for the original constrained problem, then there exists λ^* such that (λ^*, x^*) is a stationary point for the Lagrange function

This equality implies that the minimizer of f is also a minimizer of L .

If we apply FONC to L at x^* , we have

If we obtain a minimizer of L , it may be a minimizer of f also. That is, not all stationary points yield a solution of the original problem. Thus, the method of Lagrange multipliers yields a necessary condition for optimality in constrained problems, but not sufficient.

Finally, Lagrange conditions can be stated as follows:

where the set $\{\dots\dots\dots\}$ is linearly independent.

*** A point x^* satisfying (1) and (2) need not to be an extremizer.***

These conditions are called first order conditions.

(1) $\rightarrow$

(2) $\rightarrow$

Economic Interpretation of lagrange multipliers

If the RHS of constraint i is increased by α_i , then the optimal solution of the problem will depend on α_i also.

The value of the lagrange multiplier at the solution of the problem is equal to the rate of change (contribution) in the value of the objective function as the constrained relaxed.

If $\lambda_i = 0$, then the corresponding constraint is ineffective. The larger λ_i corresponds more effective constraints.

Second Order Conditions

We assume that

- $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $g: \mathbb{R}^n \rightarrow \mathbb{R}^k$ are twice continuously differentiable functions in $\mathbb{R}^n$
- the set $\{ \nabla g_i(x^*) \mid i=1, \dots, k \}$ is linearly independent
- the feasible direction set is defined as $H(x) = \{ d \in \mathbb{R}^n \mid \nabla g_i(x)^T d = 0, i=1, \dots, k \}$

Then the second order (sufficient) condition for a local minimizer is that $\nabla^2 f(x^*)$ is positive definite over $H(x^*)$.

2. Problems with equality and inequality constraints

We consider the following problem

For the above general problem, we adopt the following definitions.

Definition: An inequality constraint $g_j(x) \leq 0$ is said to be **active** at x^* if $g_j(x^*) = 0$. It is inactive at x^* if $g_j(x) < 0$.

By convention, we consider an equality constraint to be always active.

Definition: Let x^* satisfies $g_i(x^*) = 0$ for each i and $g_j(x^*) \leq 0$. And let $J(x^*)$ be the index set of active inequality constraints, that is,

We again assumed that the vectors in
are linearly independent.

First order necessary condition (Karush – Kuhn – Tucker (KKT) Condition)

f, g_i and g_j are continuously differentiable functions in R^n . Let x^* be a minimizer for the problem of minimizing f subject to $g_i(x) = 0$ and $g_j(x) \leq 0$. Then the Lagrangian of the problem is:

Nonlinear Constrained Optimization

3. Problems under the nonnegativity constraints

Consider the following problem.

Min $f(x)$

subject to $x_i \geq 0$; $i=1,2,\dots,n$.

With the following arrangement

$g_i(x)$: $-x_i \leq 0$; $i=1,2,\dots,n$

the above problem can be converted to the following form which is a problem with inequality constraints (Title 2)

Min $f(x)$

s.t. $g_i(x) \leq 0$; $i=1,2,\dots,n$

For the above problem, we adopt the FONC and SOSC in title 2:

Observe that $L(x,\mu)=$

FONC:

We search for points satisfying (1)-(4), and treat these points as candidate minimizers.

(1)

Considering the above equality, the last form of FONC can be constructed as follows:

(5)→

(6)→

(7)→

We have n equations to find n unknowns (n variables x in R^n). Observe that the candidate points must satisfy (5) and (7).

SOSC:

Then the SOSC for x^* to be a minimizer is that $\nabla^2 f(x^*)$ is positive definite over $H(x^*)$, that is for

Similarly, the SOSC for x^* to be maximizer is that $\nabla^2 f(x^*)$ is negative definite over $H(x^*)$.

Example1: $f(x, y, z) = 2x^2 + 3y^2 + z^2 - 2xy + 2xz - 2x - 4z$

$$x \geq 0, \quad y \geq 0, \quad z \geq 0$$

Example2:

$$f(x) = x_1^2 + x_1x_2 + 2x_2^2 - 6x_1 - 14x_2$$

$$\text{s.t.} \quad x_1 + x_2 + x_3 = 2$$

$$-x_1 + 2x_2 \leq 3$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

Example3:

$$f(x) = x_1^2 + x_2^2$$

$$\text{s.t.} \quad x_1^2 + 2x_2^2 - 1 = 0$$